

OECD Artificial Intelligence Papers

Competition in the age of AI

Initial evidence from microdata

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Competition in the age of AI: initial evidence from microdata



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Abstract

Artificial intelligence (AI) can reshape markets, yet its implications for competition remain underexplored. This paper develops a conceptual framework distinguishing AI users from developers, and generative (GenAI) from non-generative AI, showing how competitive mechanisms differ across these dimensions. Using multiple microdata sources, the analysis documents several findings. Adoption of non-GenAI is not associated with significant increases in market power. Descriptive evidence on firms' exposure to GenAI suggests opportunities for smaller firms alongside advantages for firms with stronger existing capabilities. Concentration in AI innovation is correlated with higher sales concentration. AI-related patenting is associated with faster markup growth, particularly in the ICT sector, where AI is an output. The AI start-up ecosystem is dynamic and attracts substantial venture capital, but start-ups are frequently acquired by large incumbents. Overall, the evidence points to a dynamic yet uneven landscape, underscoring the need for continued monitoring as AI diffusion progresses.

Résumé

L'intelligence artificielle (IA) peut remodeler les marchés, mais ses implications pour la concurrence demeurent peu étudiées. Ce document élabore un cadre conceptuel distinguant les utilisateurs d'IA de ses développeurs, ainsi que l'IA générative (GenAI) de l'IA non générative, en montrant comment les mécanismes concurrentiels diffèrent selon ces dimensions. À partir de multiples sources de microdonnées, l'analyse établit plusieurs constats. L'adoption de l'IA non générative n'est pas associée à des hausses significatives du pouvoir de marché. Les données descriptives sur l'exposition des entreprises à la GenAI suggèrent des opportunités pour les petites entreprises, ainsi que des avantages pour celles dont les capacités existantes sont plus solides. La concentration de l'innovation en IA est corrélée à une concentration plus élevée des ventes. L'activité de brevetage liée à l'IA est associée à une croissance plus rapide des marges, en particulier dans le secteur des TIC, où l'IA constitue un produit. L'écosystème des start-ups en IA est dynamique et attire des capitaux-risque substantiels, mais ces start-ups sont fréquemment rachetées par de grands acteurs établis. Dans l'ensemble, les données dépeignent un paysage dynamique mais inégal, soulignant la nécessité d'un suivi continu à mesure que la diffusion de l'IA progresse.

Kurzfassung

Künstliche Intelligenz (KI) kann Märkte umgestalten, doch ihre Auswirkungen auf den Wettbewerb sind nach wie vor wenig erforscht. Die vorliegende Studie entwickelt einen konzeptionellen Rahmen, der zwischen KI-Nutzern und KI-Entwicklern sowie zwischen generativer KI (GenAI) und nicht-generativer KI differenziert, und zeigt auf, wie sich die Wettbewerbsmechanismen entlang dieser Dimensionen unterscheiden. Auf Grundlage verschiedener Mikrodatenquellen dokumentiert die Analyse mehrere Befunde. Die Einführung nicht-generativer KI geht nicht mit einem signifikanten Anstieg der Marktmacht einher. Eine deskriptive Betrachtung der Exposition von Unternehmen gegenüber GenAI deutet auf Chancen für kleinere Unternehmen sowie auf Vorteile für solche mit ausgeprägteren bestehenden Fähigkeiten hin. Die Konzentration der KI-Innovation korreliert mit einer höheren Umsatzkonzentration. KI-bezogene Patentaktivität ist mit einem rascheren Wachstum der Preisaufschläge verbunden, insbesondere im IKT-Sektor, in dem KI ein Produkt darstellt. Das Start-up-Ökosystem der KI ist dynamisch und zieht erhebliches Wagniskapital an, doch werden diese Start-ups häufig von großen etablierten Unternehmen übernommen. Insgesamt zeichnen die Ergebnisse ein dynamisches, aber uneinheitliches Bild und unterstreichen die Notwendigkeit einer fortlaufenden Beobachtung im Zuge der weiteren Verbreitung von KI.

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Executive summary

AI has the potential to become a general-purpose technology, radically reshaping how firms organise production and compete. While a growing body of research analyses AI adoption and its impacts on productivity and labour markets, the relationship between AI and competition – especially beyond the foundation models market – remains largely unexplored. This report bridges this gap by outlining a novel conceptual framework and providing initial empirical evidence on the links between AI use, AI development and competition.

The proposed conceptual framework distinguishes between AI users (firms leveraging AI in their processes) and AI developers (firms building AI systems) and further differentiates between generative AI (GenAI) and non-generative AI (non-GenAI). These two distinctions are crucial, as actors' economic characteristics and competition concerns vary significantly. For instance, the development of foundation models is characterised by high fixed costs and economies of scale, which may undermine competition in the long run. The downstream use of these models is widely accessible to firms, creating potential for disruption via market entry. However, advantages in the ability to use them effectively might also strengthen the position of incumbents.

Empirically, the analysis focuses on competition dynamics beyond the foundation models and AI-inputs markets. It leverages a rich set of cross-country microdata, including official firm-level ICT surveys for France and Portugal, linked employer-employee data (LEED) for Portugal, alongside global AI-related patent filings and start-up data, documenting several key findings:

1. On average over the period analysed (2011-2022), focusing on France and Portugal, the use of non-GenAI does not appear associated with a systematic increase in proxies of market power. In particular, while AI adopters are generally larger and more productive, their markups are not substantially higher than those of non-users.
2. Combining occupational data at firm level with occupation-level measures of AI exposure allows the computation of a firm-specific indicator of AI exposure which – interpreted as a proxy for AI's potential adoption – suggests that GenAI can offer opportunities for smaller and younger firms to enter the market and challenge incumbents. However, the benefits of GenAI appear more easily unlocked by firms with higher existing capabilities, such as human capital and complementary assets, posing a risk of widening productivity gaps.
3. Focusing more closely on development, without distinguishing between GenAI and non-GenAI, patent data uncover a positive correlation between concentration in AI innovation and market concentration. Concentration in AI innovation is computed as the share of AI patents accounted for by the 4 largest patentees relative to the total number of AI patents in each year, while market concentration is defined at the level of a 3-digit industry, the geographic scope of competition (domestic, European, or global), and year. The development of the technology currently mirrors existing heterogeneities among firms, potentially entrenching the advantages of established leaders. Furthermore, in ICT sectors, firms with an AI-related patent exhibit higher markup growth, suggesting that owning AI technology may contribute to market power, but only in those sectors where AI is an output rather than an input.

4. The AI entrepreneurial ecosystem remains highly dynamic, characterised by sustained entry and significant venture capital investments, particularly in GenAI. However, this is accompanied by intense acquisition activity by large incumbents. While acquisitions can favour entry of start-ups in the market, facilitate their exit, and transfer technology, they also raise concerns about potential “killer acquisitions” that may stifle future innovation and growth prospects.

Overall, the early empirical evidence documented presents a mixed picture. The current ecosystem displays signs of both dynamism and potential emerging concentration risks. However, as AI diffusion is still in its early stages (20.2% of OECD enterprises in 2025, according to OECD ICT usage data), it may take time before competition dynamics become fully observable. These findings underscore the need for continuous and vigilant monitoring of AI markets to ensure that the technology fosters innovation and contestability, rather than consolidating dominance. Finally, the report highlights the need for more comprehensive data collection on AI use and development, which is crucial for effectively monitoring the relationship between AI and competition.

Synthèse

L'IA pourrait devenir une technologie à usage général, transformant radicalement la manière dont les entreprises organisent leur production et se font concurrence. Si un nombre croissant de travaux analysent l'adoption de l'IA et ses effets sur la productivité et le marché du travail, la relation entre l'IA et la concurrence – en particulier au-delà du marché des modèles de fondation – demeure largement inexplorée. Ce rapport comble cette lacune en exposant un cadre conceptuel inédit et en fournissant des données empiriques initiales sur les liens entre l'utilisation de l'IA, son développement et la concurrence.

Le cadre conceptuel proposé distingue les utilisateurs d'IA (entreprises qui mobilisent l'IA dans leurs processus) des développeurs d'IA (entreprises qui conçoivent des systèmes d'IA), et différencie en outre l'IA générative (GenAI) de l'IA non générative (non-GenAI). Ces deux distinctions sont essentielles, car les caractéristiques économiques des acteurs et les préoccupations en matière de concurrence varient sensiblement. Ainsi, le développement des modèles de fondation se caractérise par des coûts fixes élevés et des économies d'échelle, qui pourraient nuire à la concurrence à long terme. L'utilisation en aval de ces modèles est largement accessible aux entreprises, ce qui crée des opportunités d'entrée sur le marché pour de nouveaux acteurs. Toutefois, les avantages liés à la capacité de les utiliser efficacement pourraient également renforcer la position des acteurs établis.

Sur le plan empirique, l'analyse porte sur les dynamiques concurrentielles au-delà des marchés des modèles de fondation et des intrants de l'IA. Elle s'appuie sur un riche ensemble de microdonnées multi-pays, dont les enquêtes officielles sur les TIC au niveau des entreprises en France et au Portugal, des données employeurs-salariés appariées (LEED) pour le Portugal, ainsi que des dépôts mondiaux de brevets liés à l'IA et des données sur les start-ups. L'analyse fait ressortir plusieurs constats principaux:

5. En moyenne sur la période analysée (2011-2022), en France et au Portugal, l'utilisation de l'IA non générative ne semble pas associée à une augmentation systématique des indicateurs indirects du pouvoir de marché. En particulier, si les entreprises adoptantes sont généralement plus grandes et plus productives, leurs marges ne sont pas sensiblement supérieures à celles des non-utilisatrices.
6. La combinaison de données professionnelles au niveau des entreprises avec des mesures d'exposition à l'IA par profession permet de calculer un indicateur d'exposition à l'IA propre à chaque entreprise, qui peut être interprété comme une mesure indirecte de l'adoption potentielle de l'IA. Cet indicateur suggère que la GenAI peut offrir aux entreprises plus petites et plus jeunes des occasions d'entrer sur le marché et de faire concurrence aux acteurs établis. Toutefois, les bénéfices de la GenAI semblent plus facilement être captés par les entreprises dotées de capacités existantes plus solides, telles que le capital humain et les actifs complémentaires, faisant peser un risque de creusement des écarts de productivité.
7. En s'intéressant de plus près au développement, sans distinguer entre la GenAI et l'IA non générative, les données de brevets révèlent une corrélation positive entre la concentration de l'innovation en IA et la concentration des marchés. La concentration de l'innovation en IA est mesurée par la part des brevets d'IA détenue par les quatre plus grands déposants rapportée au nombre total de brevets d'IA pour chaque année, tandis que la concentration des marchés est

définie au niveau d'une branche à trois chiffres, du champ géographique de la concurrence (national, européen ou mondial) et de l'année. Le développement de la technologie reflète actuellement les disparités préexistantes entre les entreprises, ce qui pourrait consolider les avantages des acteurs établis. Par ailleurs, dans les secteurs des TIC, les entreprises détenant un brevet lié à l'IA affichent une croissance plus forte de leurs marges, ce qui donne à penser que la détention d'une technologie d'IA peut contribuer au pouvoir de marché, mais uniquement dans les secteurs où l'IA constitue un produit plutôt qu'un intrant.

8. L'écosystème entrepreneurial de l'IA demeure très dynamique, marqué par une entrée soutenue et des investissements en capital-risque importants, en particulier dans la GenAI. Cette dynamique s'accompagne toutefois d'une intense activité de rachat par de grands acteurs établis. Si les acquisitions peuvent favoriser l'entrée des start-ups sur le marché, faciliter leur sortie et permettre des transferts de technologie, elles soulèvent également des préoccupations quant à d'éventuelles « acquisitions prédatrices » susceptibles d'étouffer l'innovation et les perspectives de croissance futures.

Dans l'ensemble, les premières données empiriques recueillies dressent un tableau contrasté. L'écosystème actuel présente des signes à la fois de dynamisme et de risques potentiels de concentration émergente. Cependant, la diffusion de l'IA en étant encore à ses débuts (20,2 % des entreprises de l'OCDE en 2025, selon les données de l'OCDE sur l'usage des TIC), il pourrait s'écouler un certain temps avant que les dynamiques concurrentielles ne deviennent pleinement observables. Ces constats soulignent la nécessité d'un suivi continu et vigilant des marchés de l'IA, afin de garantir que cette technologie favorise l'innovation et la contestabilité plutôt que la consolidation de positions dominantes. Enfin, le rapport met en évidence la nécessité d'une collecte de données plus complète sur l'utilisation et le développement de l'IA, indispensable pour suivre efficacement la relation entre l'IA et la concurrence.

Zusammenfassung

KI könnte sich zu einer Allzwecktechnologie entwickeln und die Art und Weise, wie Unternehmen ihre Produktion organisieren und im Wettbewerb stehen, grundlegend verändern. Während eine wachsende Zahl von Arbeiten die Einführung von KI und ihre Auswirkungen auf Produktivität und Arbeitsmärkte untersucht, ist das Verhältnis zwischen KI und Wettbewerb – insbesondere jenseits des Marktes für Basismodelle – weitgehend unerforscht. Dieser Bericht schließt diese Lücke, indem er einen neuartigen konzeptionellen Rahmen darlegt und erste empirische Befunde zu den Zusammenhängen zwischen KI-Nutzung, KI-Entwicklung und Wettbewerb liefert.

Der vorgeschlagene konzeptionelle Rahmen unterscheidet zwischen KI-Nutzern (Unternehmen, die KI in ihren Prozessen einsetzen) und KI-Entwicklern (Unternehmen, die KI-Systeme erstellen) und differenziert darüber hinaus zwischen generativer KI (GenAI) und nicht-generativer KI (non-GenAI). Diese beiden Unterscheidungen sind von zentraler Bedeutung, da sich die wirtschaftlichen Merkmale der Akteure und die wettbewerbsrechtlichen Bedenken erheblich unterscheiden. So ist die Entwicklung von Basismodellen durch hohe Fixkosten und Skaleneffekte gekennzeichnet, die den Wettbewerb langfristig beeinträchtigen könnten. Die nachgelagerte Nutzung dieser Modelle ist für Unternehmen weithin zugänglich und schafft Potenzial für disruptive Effekte durch Markteintritt. Vorteile bei der Fähigkeit, sie wirksam einzusetzen, könnten jedoch auch die Position etablierter Unternehmen stärken.

Empirisch konzentriert sich die Analyse auf Wettbewerbsdynamiken jenseits der Märkte für Basismodelle und KI-Vorleistungen. Sie stützt sich auf einen umfangreichen Bestand länderübergreifender Mikrodaten, darunter amtliche IKT-Erhebungen auf Unternehmensebene für Frankreich und Portugal, verknüpfte Arbeitgeber-Arbeitnehmer-Daten (LEED) für Portugal sowie weltweite KI-bezogene Patentanmeldungen und Start-up-Daten, und dokumentiert mehrere zentrale Befunde:

1. Für Frankreich und Portugal scheint die Nutzung nicht-generativer KI im Durchschnitt des Analysezeitraums (2011-2022) nicht mit einem systematischen Anstieg der Näherungsindikatoren für Marktmacht verbunden zu sein. Insbesondere sind die anwendenden Unternehmen zwar in der Regel größer und produktiver, ihre Preisaufschläge fallen jedoch nicht wesentlich höher aus als jene der Nichtnutzer.
2. Die Verknüpfung von Beschäftigungsdaten auf Unternehmensebene mit berufsbezogenen Maßen der KI-Exposition ermöglicht die Berechnung eines unternehmensspezifischen Indikators der KI-Exposition, der – als Näherungswert für die potenzielle Einführung von KI verstanden – darauf hindeutet, dass GenAI kleineren und jüngeren Unternehmen Chancen bieten kann, in den Markt einzutreten und etablierte Unternehmen herauszufordern. Die Vorteile von GenAI scheinen sich jedoch leichter von Unternehmen mit ausgeprägteren bestehenden Fähigkeiten – etwa Humankapital und komplementären Vermögenswerten – erschließen zu lassen, was das Risiko einer Vergrößerung der Produktivitätsunterschiede birgt.
3. Bei näherer Betrachtung der Entwicklung, ohne Unterscheidung zwischen GenAI und nicht-generativer KI, zeigen die Patentdaten eine positive Korrelation zwischen der Konzentration der KI-Innovation und der Marktkonzentration. Die Konzentration der KI-Innovation wird als Anteil der vier größten Patentanmelder an der Gesamtzahl der KI-Patente im jeweiligen Jahr berechnet; die

Marktkonzentration wird auf Ebene der dreistelligen Branche, des geografischen Wettbewerbsraums (national, europäisch oder global) und des Jahres definiert. Die Entwicklung der Technologie spiegelt derzeit bestehende Heterogenitäten zwischen Unternehmen wider und könnte die Vorteile etablierter Marktführer verfestigen. Darüber hinaus weisen im IKT-Sektor Unternehmen mit einem KI-bezogenen Patent ein höheres Wachstum ihrer Preisaufschläge auf, was darauf hindeutet, dass der Besitz einer KI-Technologie zur Marktmacht beitragen kann, allerdings nur in jenen Sektoren, in denen KI ein Produkt und keine Vorleistung darstellt.

4. Das unternehmerische Ökosystem der KI bleibt äußerst dynamisch und ist durch anhaltende Markteintritte und erhebliche Wagniskapitalinvestitionen gekennzeichnet, insbesondere im Bereich der GenAI. Damit geht jedoch eine intensive Übernahmetätigkeit großer etablierter Unternehmen einher. Zwar können Übernahmen den Markteintritt von Start-ups begünstigen, deren Ausstieg erleichtern und Technologietransfer ermöglichen, doch werfen sie auch Bedenken hinsichtlich möglicher „Killer-Übernahmen“ auf, die künftige Innovationen und Wachstumsaussichten hemmen könnten.

Insgesamt zeichnen die ersten dokumentierten empirischen Befunde ein gemischtes Bild. Das gegenwärtige Ökosystem weist Anzeichen sowohl von Dynamik als auch von potenziellen aufkommenden Konzentrationsrisiken auf. Da sich die Verbreitung von KI jedoch noch in einem frühen Stadium befindet (20,2 % der Unternehmen im OECD-Raum nutzen im Jahr 2025 KI, gemäß den OECD-Daten zur IKT-Nutzung), könnte es einige Zeit dauern, bis die Wettbewerbsdynamiken vollständig sichtbar werden. Diese Befunde unterstreichen die Notwendigkeit einer kontinuierlichen und wachsamten Beobachtung der KI-Märkte, um sicherzustellen, dass die Technologie Innovation und Bestreitbarkeit fördert, statt marktbeherrschende Stellungen zu festigen. Abschließend hebt der Bericht die Notwendigkeit einer umfassenderen Datenerhebung zur Nutzung und Entwicklung von KI hervor, die für eine wirksame Beobachtung des Verhältnisses zwischen KI und Wettbewerb von entscheidender Bedeutung ist.

1 Introduction

Artificial Intelligence (AI) has the potential to radically transform economies by reshaping how people work and how firms organise their production processes. Due to its broad applicability, AI holds the potential of becoming a general-purpose technology (Calvino, Haerle and Liu, 2025^[1]), stimulating an extensive debate among academics and policymakers on its benefits and risks (OECD, 2024^[2]; Korinek, 2024^[3]).¹ A growing body of research has analysed AI adoption across the economy (Calvino and Fontanelli, 2023^[4]), and its potential impact on several economic dimensions, including labour market (Acemoglu et al., 2022^[5]) and aggregate productivity (e.g. Filippucci et al. (2024^[6]); Acemoglu (2024^[7]); and Aghion and Bunel (2024^[8])). Little is known, however, about the relationship between AI use and development and competition, with the overall discussion remaining at an exploratory stage (Kowalski, Volpin and Zombori, 2024^[9]).

At the same time, a lively debate has emerged on competition trends across OECD Member countries. Evidence shows rising concentration and markups (Calligaris et al., 2024^[10]; Bajgar et al., 2023^[11]) across a broad set of economies, together with declining entry and job reallocation rates (Calvino, Criscuolo and Verlhac, 2020^[12]).² While these trends can be associated with a decline in competition, for example due to excessive regulatory barriers (Covarrubias, Gutiérrez and Philippon, 2019^[13]), they might also reflect competition in action (Shapiro and Yurukoglu, 2024^[14]; Miller, 2025^[15]). Investments in intangibles such as software, research and development (R&D), data, and marketing, have become increasingly important; this shift may have benefited larger firms, as they have the financial resources to make the initial investment and the scale to fully benefit from intangible assets (De Ridder, 2024^[16]; Autor et al., 2020^[17]; Bessen et al., 2020^[18]). As a result, larger firms might have been able to grow faster and offer better or cheaper products through increased efficiency. In the short run, therefore, rising industry concentration, may not necessarily reflect weaker competition, but rather higher aggregate productivity among leading firms. In the long run, however, industries with high concentration and high markups might become more entrenched and less dynamic, with several potential concerns on long run economic growth, incentives to innovation and consumer welfare (Aghion et al., 2023^[19]).³

AI technologies share a few characteristics with intangible assets, as well as very specific features, that may create significant entry barriers.⁴ These include substantial upfront and sunk investments, which are more challenging for young and small firms to secure compared to established companies. Additionally, they can require vast amounts of data to be efficient and effective, which are often accessible only to incumbent firms. All these features can act as an entry barrier and contribute to widening gaps between firms.

Given AI's potential impact on competition and business dynamism, monitoring its ecosystem is crucial. Competitive dynamics can differ substantially across the AI value chain – between upstream AI developers, i.e. who design and train foundation models, and downstream developers and users, i.e. who adapt and apply these systems to create products and services for specific markets. They also interact with the type of technology involved – such as single-purpose AI versus general-purpose AI. These dimensions influence competition in distinct ways, shaping who benefits, how innovation spreads, and how markets evolve.

Recently, several authors have characterised the AI ecosystem as a stack composed by different layers: hardware, which includes computing chips and specialised software; cloud platforms offering computing services that are crucial for model development and deployment; data sources; foundation models; AI

applications making use of the models such as virtual assistant or autonomous driving applications (Hagiwara and Wright, 2025^[20]; Narechania and Sitaraman, 2024^[21]; Gambacorta and Shreeti, 2025^[22]; OECD, 2025^[23]). This work focuses on the downstream parts of the stack, without considering issues, for example, in the chip production layer, nor specifically focusing on competition in the market for foundation models.

In particular, this work defines a conceptual framework capturing these distinctions to study the impact of AI on competition from a firm dynamics perspective.⁵ It then empirically characterises – based on the early wave of AI diffusion – the links between AI and different proxies of competition, such as industry concentration and firm markups, relating the initial empirical evidence to the conceptual framework.⁶

First, the framework distinguishes between non-generative AI (non-GenAI) and generative AI (GenAI) tools. Before the commercial release of GenAI tools such as ChatGPT or DALL-E 2 in late 2022, most AI applications relied on non-generative machine learning (ML) models. These tools learn from data, identify patterns, and make predictions, typically applied to specific tasks.⁷ Non-GenAI share some characteristics with intangible assets, as their adoption often involves upfront costs, including investments in complementary assets, like ICT infrastructure and workforce training (Calvino and Fontanelli, 2023^[4]). These features, combined with the ownership of the data used to train and apply these tools, suggest a significant role of economies of scale as potential entry barriers, possibly leading to concentration in the markets where firms adopt AI. However, some AI tools can also be obtained from cloud service providers, such as Amazon Web Services, Microsoft Azure, or Google Cloud, which provide data analysis software, together with customisable AI solutions. These services reduce the need for the complementary infrastructure, and the associated fixed costs become, to some extent, less relevant. Although the standardisation of the offered services increases the pool of potential AI users, even the easiest applications may require substantial human capital, making AI adoption potentially uneven across firms. The productivity gains enabled by AI, by lowering firms' marginal costs, might allow them to charge higher markups while lowering prices (depending on cost-price pass-through), thereby leading to larger market shares. AI's impact on competition also extends beyond productivity gains and cost reduction. Firms that adopt AI technologies can also introduce innovative or more tailored products or services, resulting in an increase in firms' market power. These factors, alongside productivity gains, can lead to higher markups and potentially consolidate market power.

Unlike non-generative ones, GenAI tools can create new content, such as text, images, or video, that mimics human outputs. GenAI rely on foundation models (Bommasani et al., 2021^[24]) – AI models built using deep learning methods and trained on vast datasets.⁸ Foundation models are versatile in their applications and can enhance business processes across several dimensions.⁹ GenAI applications can substitute human inputs in certain tasks while complementing or augmenting them in others. Relevantly, much of GenAI's potential lies in its ability to integrate with other software (such as office suites, programming tools like in the case of GitHub Copilot, or business analytics programmes). These features, helping reduce costs, can accelerate firms' growth and improve productivity.¹⁰ When used by established firms, which may be more prone to adoption or to leverage its full potential through organisational change, GenAI might reinforce their position in the market. At the same time, it might drive new business models, enabling young firms to enter new markets and challenge more conservative incumbents.

Second, the framework distinguishes between AI users and AI developers, although the line between the two is not always clear-cut.¹¹ Users can be thought of as firms purchasing AI software to apply it to specific tasks within their core operations. For example, they include firms that use anomaly detection algorithms for quality controls (non-GenAI) or firms that subscribe to an AI-powered assistant integrated into their management software (GenAI). AI developers, on the other hand, are firms that engage in the development of AI systems or AI-based solutions for commercial purposes. Consequently, the framework classifies firms developing AI solely for internal use as users, as their competitive dynamics are driven by the application of AI to their core business rather than by the supply of AI tools.

AI developers can be divided into two main categories. The first includes developers of frontier foundation models – large models that power GenAI tools such as ChatGPT or Google Gemini. Some authors argue that the market of frontier foundation models is inherently inclined towards concentration, although with fierce competition among developers of frontier models. Training and maintaining such models require high fixed costs and vast data and computing, potentially restricting entry to a few well-funded firms. Further concerns about competition arise from the potential for integration with other firms in the AI stack (Korinek and Vipra, 2024^[25]; Aguirre, 2024^[26]; Hagiú and Wright, 2025^[20]; André et al., 2025^[27]; OECD, 2025^[23]).¹² The second category is more heterogeneous. It includes developers of smaller, task-specific GenAI models designed for ease of deployment and lower resource use. Some of these derive from larger foundation models, while others are independently built by small firms.^{13,14} A further group includes non-GenAI developers, which can be relatively large, and offer standardised solutions often bundled with cloud services, such as Microsoft's Azure, which provide data science tools to its users. Non-GenAI developers also include other ICT firms, not necessarily large, and providing more tailored solutions. A common characteristic of small providers of AI services is their tendency to rely on specialised workers. Moreover, such actors tend to rely on cloud services, often combined with the client's own infrastructure, rather than the deployment of a proprietary one. These features allow such firms to be more cost-effective when providing tailored solutions that reflect customer preferences and the application's specific requirements.¹⁵ Finally, companies can also decide to autonomously develop their AI solution, often relying on internal human capital and targeting internal problems.¹⁶ The dependence of small AI firms on larger cloud providers might pose some challenges from a competition perspective due to the presence of switching costs and potential monopolistic power. Combining these dimensions, the framework identifies four categories (GenAI users, GenAI developers, non-GenAI users, and non-GenAI developers) and helps identify how AI relates to competition. By adopting a firm dynamics perspective, this work complements other streams of work studying algorithmic pricing and the role of AI in facilitating collusion (Calvano et al., 2020^[28]; OECD, 2025^[29]).

After outlining the conceptual framework, this work provides initial empirical insights into the link between AI and competition, relating, as much as possible, the empirical evidence with the characterisation of the AI ecosystem provided by the conceptual framework.¹⁷

The initial empirical evidence can be summarised as follows. First, analysis of firm-level ICT surveys – more closely related to non-GenAI users – suggests that, on average and in the period under analysis, AI use by firms does not seem related to substantial increases in the examined proxies of market power in France and Portugal. This is particularly true for markups trends, which do not appear significantly different between AI users and other firms, while for market shares dynamics some specifications point to a possible role of AI strengthening the edge of most competitive firms.

Second, focusing more closely on potential GenAI users, leveraging information about AI exposure based on the occupations of workers within firms, analysis of Portuguese Linked Employer-Employee Data (LEED) suggests that on the one hand generative AI may represent a relevant opportunity for smaller and younger firms, which may be well positioned to adopt AI and challenge market leaders.¹⁸ On the other hand, benefits of GenAI may be unlocked more easily by firms that have higher productivity and capabilities. This may highlight risks of increasing divides between the most productive firms and the rest of the business population.

Third, the report uses data on patenting activities to proxy for AI development which are jointly analysed with proxies of market power.¹⁹ The analysis reports mixed findings on the global concentration of AI-related patenting activity. After a significant increase in concentration levels, which is evident after 2011 – often considered a turning point in AI developments –, a decreasing trend can be observed thereafter. An overall long run decline in AI patenting concentration appears also evident when considering a longer time span (i.e., a long difference between 2001 and 2021). Such a decline in concentration seems to be driven mostly by an increasing number of medium-sized innovators rather than an absolute decline in the activity

of the largest players. At the same time, the analysis also shows a positive correlation between concentration in AI patent filings and market concentration. On the one hand, market concentration may be reinforced by the advantages brought in by AI patenting by market leaders; on the other hand, leaders in concentrated industries might have the scale or position needed to invest in AI innovation. Finally, firms with AI patenting activities have higher markup growth rates, a trend driven mostly by the ICT sectors. Within these sectors, firms associated with AI-related patents have higher markups. This might suggest an effective role of AI in creating market power, but only in those sectors where AI is an output rather than an input.

The empirical section concludes by examining the AI-related entrepreneurial ecosystem. Leveraging data from the OECD Start-up Database and applying a Large Language Model (LLM) to classify start-ups into GenAI, non-GenAI, and non-AI, the analysis reveals a highly dynamic environment. This ecosystem is characterised by the sustained entry of AI-related start-ups in recent years and substantial volumes of venture capital (VC) funding directed toward these entities, especially for GenAI start-ups. Entry and capital availability are essential elements of a vibrant entrepreneurial ecosystem that can, over the medium to long run, challenge today's incumbents. However, the analysis also documents intense acquisition activity of AI start-ups – mainly those active in GenAI – by incumbent firms, raising concerns about potential killer acquisitions that may stifle future innovation potential and growth prospects (Berger et al., 2025^[30]).

The relationship between AI diffusion and competition is complex and this report contributes to the discussion by providing an initial exploration of some of its facets. Overall, this paper provides a mixed, rather than an alarming, picture. These findings must be interpreted with caution for different reasons. First, this analysis has leveraged imperfect proxies of AI adoption, development, and competition. Moreover, while relating the findings to the presented conceptual framework, data limitations required certain simplifications. Given the recent surge in generative AI activity, existing datasets do not yet fully capture the phenomenon. Finally, from an economic viewpoint, the findings represent a short run perspective, depicting an ecosystem still at early stages of competition, where firms' uptake of AI remains relatively low with respect to other digital technologies. As often happens, potential competition issues may take time to materialise, with first-mover firms consolidating their market power in the long run. For these reasons, these conclusions are preliminary. The relationship between AI and market power requires continuous and vigilant monitoring, and new trends may emerge as the technology matures and more comprehensive data become available. Given its broad perspective, the current framework and empirical evidence can serve as a tool and initial basis for further monitoring the evolution of the AI and competition landscapes.

The remainder of this paper is structured as follows. Section 2 presents the conceptual framework and briefly reviews the relevant literature. Section 3 describes the data sources. Section 4 outlines the empirical strategy, presents the main results and discusses the findings. Section 5 concludes.

2 The conceptual framework

This section provides a detailed description of the conceptual framework, summarised in Table 2.1. The table gathers the different categories of the framework (generative vs. non-generative AI and users vs. developers of AI) and provides a brief description of the economic characteristics of each category, together with the potential implications for the competitive environment. Each category is separately discussed further below, also in relation to the relevant literature.

While comprehensive in several respects, notably in relation to the aim of broadening the scope of discussion beyond a single type of AI user or technological use, some relevant caveats should be kept in mind in relation to the proposed framework.²⁰ As previously mentioned, the distinctions between quadrants may be sometimes blurred, and competition dynamics in AI markets are interdependent with those along the broader technology stack and in the value chain, which are elements not explicitly embedded in the scope of the current analysis.

Generative AI users

The top left quadrant describes the characteristics of GenAI users, namely firms that adopt GenAI tools in their production processes and related activities. These tools can generate new content and support a wide range of time-consuming tasks (e.g. email drafting, customer service, logistics, sales analysis, product innovation, legal and risk analysis, and for coding and software development). Users are heterogeneous, ranging from non-sophisticated ones, that mostly use easily accessible tools such as ChatGPT, but also more sophisticated ones who fine-tune LLM for their own use.

The diffusion of GenAI tools may potentially display pro-competitive effects, for example by accelerating product development, improving the identification of market gaps, and enhancing efficiency, potentially forcing less productive firms to exit the market and fostering business dynamism (OECD, 2025^[29]). At the same time, greater scope for product differentiation, quality improvements and higher productivity may allow some firms to sustain higher markups and expand market shares.²¹ This might reflect the ability of firms to outperform their competitors by establishing better business strategies. Uneven adoption, however, raises competition concerns. Small firms might have a lower absorptive capacity, due to limited resources, difficulties in identifying effective use cases, or insufficient complementary investments and organisational change. As a result, gaps between leading and lagging firms may widen. Furthermore, in the long run, incumbents may leverage their advantages to entrench their position and exert market power.

The evidence on the implications of adopting GenAI tools for competition is quite limited, which is unsurprising given that the first commercial applications were released after 2020. However, a growing body of studies explores the potential uses and impacts of GenAI in a business context. While many focus on labour market impacts (OECD, 2025^[31]) or on the effects for productivity, innovation and entrepreneurship leveraging experimental settings (see Calvino, Reijerink and Samek (2025^[32]) for a recent literature review), they also provide some insights into the potential influence on competition. The evidence suggests that GenAI adoption operates partly through mechanisms of complementarity with other inputs, including labour and capital (e.g. when integrated into software), while in some tasks, particularly clerical work, GenAI may be a substitute for human labour. From a competition perspective, these

mechanisms alter firms' cost structures and barriers to entry. For instance, if GenAI enables smaller firms with fewer resources to achieve quality levels previously reserved to incumbents, it could foster competition; conversely, if effective adoption still requires substantial complementary assets, it may reinforce existing advantages.

Orchard and Tasiemski (2023^[33]) analyse the possible implications of GenAI for different aspects of business and the economy, highlighting its disruptive potential. They show how it can be leveraged to enter market segments where incumbent firms offer products that are overly refined for certain customers. The use of GenAI tools to serve these customers, although under human supervision, would increase market reach at a lower cost and higher efficiency. Cooper (2024^[34]) highlights how, despite the reasons for firms to adopt AI are manifold (improving efficiency, business agility, or productivity), significant benefits are related to AI use in product development, where GenAI tools can be used at various stages, from concept creation to the design and testing of the products. Calvino, Reijerink and Samek (2025^[32]) review experimental evidence on how GenAI affects productivity, innovation, and entrepreneurship, showing that it can deliver significant gains, but impacts vary by task design, user expertise, and trust in AI outputs. Overall, the paper argues that human-AI complementarity, rather than substitution, is key to realising the technology's economic potential while addressing its limitations.

Generative AI developers

The top-right quadrant describes the main economic features of GenAI developers, including those producing advanced foundation models that power tools, such as Chat-GPT or Google Gemini.

There is a relatively broad consensus in the literature on the characteristics of the market for foundation models (Azoulay, Krieger and Nagaraj, 2024^[35]; Gans, 2024^[36]; Marar, 2024^[37]; Schrepel and Potts, 2024^[38]; Korinek and Vipra, 2024^[25]). These models are characterised by economic and technological aspects that may lead towards concentration. At the same time, recent OECD work highlights signs of dynamism in the AI model development market. In particular, André et al (2025^[27]) provide empirical evidence on the price and quality of the main generative AI models. Their analysis shows that the market exhibits signs of dynamism, characterised by a rapid turnover of models at the frontier and a fall in quality-adjusted prices. However, they also highlight persistent risks for competition, noting that while the market is currently contestable, the concentration of critical inputs such as compute and data, alongside the presence of vertically integrated hyper-scalers, could result in lower dynamism in the future. Another recent work regarding the market of LLMs is by Demirer et al. (2025^[39]), who highlight significant dynamism, with a significant role played by open-source models.

From a conceptual viewpoint, Gans (2024^[36]) distinguishes between AI outputs (the predictions made with AI), from the different types of inputs: data (both for training and as input), talent (data scientists and developers), and compute (hardware), and highlights how there is a potential for concentration in both output and inputs. Two considerations are particularly important from the competition perspective, according to Gans (2024^[36]). First, it is possible to assign a distinct role to training data, which can be thought of as a driver of entry into a market, and to input data, which are related to costs or quality of the output of firms adopting AI.²² Second, Gans (2024^[36]) describes the mechanism leading to increasing returns to scale: data and AI attract users, whose data further improve model performance, reinforcing market power (Farboodi and Veldkamp, 2023^[40]), and giving rise to increasing returns to scale in the AI technology. This data-driven advantage can be reinforced by network effects and high switching costs. As more developers build applications upon a particular foundation model, they create a richer ecosystem of tools and integrations, increasing the platform's value for all users. Firms that embed their operations within such an ecosystem face significant switching costs, as moving to a competitor would require costly re-engineering of products and processes. Agrawal, Gans and Goldfarb (2019^[41]) point out how this

mechanism, coupled with an early adoption of its recommendation system, enabled Amazon to gain its market share.

The analysis by Gans (2024^[36]), mainly focused on the data component of AI systems, is complemented by Korinek and Vipra (2024^[25]). They describe other economic characteristics of foundation models, from the large training costs to the versatility of their application, which give rise to economies of scale and scope and, therefore, to potentially increased concentration. Moreover, the inputs required to train these models, such as computational power, skilled workers, and datasets, are becoming relatively scarcer due to high demand. As a result, access to these critical resources might become limited to a few well-funded economic entities, further concentrating market power among a small group of dominant firms. Azoulay, Krieger and Nagaraj (2024^[35]), further argue that complementary assets can act as competitive moats, reinforcing the dominant position of leading firms.

Another force highlighted by Korinek and Vipra (2024^[25]) that might lead to a weakening of competition is the natural tendency for producers of foundation models to engage in vertical integration. Upstream integrations might occur as these firms seek easier access to inputs, such as data and hardware, thereby securing their supply chains and reducing reliance on external suppliers. By controlling these critical resources, foundation model developers can lower costs, enhance quality, and ensure a more reliable supply of inputs necessary for model training. On the other hand, downstream integration may involve collaborations with or acquisitions of firms that rely on the applications of foundation models in their products or services. This strategy strengthens relationships with customers and allows for better alignment of product development with market needs, creating a more cohesive ecosystem around their technology.

Korinek and Vipra (2024^[25]) provide a picture where the market for top-tiered foundation models is difficult to enter due to the aforementioned characteristics, but characterised by strong competition among top players. They also describe a second market segment populated by developers of non-frontier models, which are designed to be lightweight, easy to deploy, and optimised for specific tasks or devices.²³ The market for non-frontier models is more competitive because it has lower requirements in terms of computational inputs, making it more accessible to a broader range of developers, and has been so far characterised by relatively swift (price) competition (Korinek and Vipra, 2024^[25]).

Non-GenAI users

The bottom-left quadrant describes users of non-GenAI. Typical applications of non-GenAI include recommender systems, anomaly detection tools, and forecasting models. A crucial aspect is whether AI will impact firms' minimum efficient scale. If adoption requires substantial investment in customised solutions, high fixed costs may restrict AI use to large firms. On the contrary, when firms can acquire AI services from cloud providers, fixed costs and minimum efficient scale are lower, with data availability for fine-tuning becoming the main constraint. Nevertheless, the eventual adoption of AI requires readiness on the management side to recognise its potential benefits, the capacity to reorganise the business processes, and the presence of skilled workers. In fact, even low levels of customisation of the AI tools acquired on demand from external providers will likely require AI developers to adapt them to the firm's needs. While the impact of AI on the labour market has received considerable attention (Acemoglu et al., 2022^[5]), the evidence on the relationship between non-GenAI and competition is rather scant due to the scarcity of granular data on AI adoption by firms.

In standard models of imperfect competition (Jaumandreu, 2024^[42]), an increase in firms' productivity is associated with an increase in their market share, with such relationship being stronger in more competitive environments.²⁴ Assuming that AI use is associated with higher productivity, the heterogeneous adoption of AI across firms can lead, in the models considered, to an increase in market share dispersion. One of the few works directly linking AI adoption by firms to industry concentration is by Babina et al. (2024^[43]),

who find that AI-investing firms experience higher sales growth, mainly associated with increased product innovation. Moreover, AI-induced growth benefits large firms in relatively concentrated industries. Most of the literature focused on the potential productivity gains from AI adoption, which can be considered one of the channels through which AI can affect competition. Among other results, Calvino and Fontanelli (2023^[4]) find that AI users tend to be, on average, more productive and larger than other firms. The productivity differentials, though, are not entirely credited to using AI but appear relevantly associated with the presence of complementary assets, such as ICT skills and digital infrastructure. Focusing on industrial AI use in the United States, McElehran et al. (2025^[44]) provide evidence of a productivity J-curve, where short-term productivity losses are followed by longer-term gains.

Analogously, rising markups, often associated with increasing sales concentration (Autor et al., 2020^[17]; De Loecker, Eeckhout and Unger, 2020^[45]), can also be attributed to an increase in firm-level productivity. Von Maydell (2024^[46]) structurally models AI as a self-improving input (through application, testing, and training) whose adoption increases firms' productivity, raises entry barriers and leads to higher concentration. A lower number of players also increases firms' market power, allowing them to increase markups to finance AI infrastructure further.

These findings highlight two issues. First, it is not only the use of AI that can affect the state of competition; the reverse is also true. Higher degrees of competition can influence how AI affects market outcomes. For instance, in highly competitive environments, efficiency gains from AI are more likely to translate into market share reallocations as firms compete on price or quality, whereas in less competitive sectors, such gains may simply be absorbed as higher margins. Second, AI use tends to be higher in large firms, which might further increase productivity and market shares dispersion. From a competition viewpoint, it is important to acknowledge the first-mover advantage that certain firms might accumulate and use to maintain their position in the market.

Varian (2019^[47]) further elicits a few possible dimensions along which AI use might impact the industrial organisation of firms and, therefore, competition. A crucial dimension involves different forms of returns to scale that can be associated with AI. First, the classical supply-side return to scale is related to minimum efficient size, and it aligns with the arguments used by De Ridder (2024^[16]) to describe the relationship between intangibles and market power. High fixed costs are required to develop AI applications, but then their use can happen at very low marginal costs, increasing firms' markups and potentially increasing their market share. The second type is learning-by-doing, intended as the quality improvement (or cost decline) due to repeated applications of AI. Usually, this is associated with the fact that AI algorithms are used for predictions and can improve through the use of new data (Farboodi and Veldkamp, 2023^[40]; Farboodi et al., 2019^[48]; Mihet and Philippon, 2019^[49]). A complementary perspective on AI-related returns to scale is provided by Lu, Phillips and Yang (2024^[50]), who treat cloud computing and AI as different technologies and study the impact of their adoption on industry dynamics in the People's Republic of China (hereafter, 'China'). Cloud computing is characterised by ease of adoption, which occurs on demand at low marginal costs and is associated with increased firm entry and exit and lower concentration (see also Caldarola and Fontanelli (2024^[51]) for evidence of cloud disproportionately benefiting smaller firms in France). AI computing is rather associated with the need for data, which advantages big firms, and with a reduction in entry rates and an increase in concentration. Another aspect considered by Varian (2019^[47]) is the possibility of extracting more value from customers through improved price discrimination techniques. A final dimension is the possibility that algorithmic pricing might lead to collusive practices (see Calvano et al. (2020^[28]) for a recent investigation), which makes supra-competitive prices sustainable.

Non-GenAI developers

The bottom-right quadrant focuses on non-GenAI developers, the supply-side counterpart of AI users, and can be broadly divided into two groups. First, there are relatively small ICT or independent AI companies,

which rely on specialised workers to write software rather than using large infrastructure. They often use cloud services, often combined with the client's infrastructure, to offer cost-effective tailored solutions reflecting customer preferences and specific application requirements. Second, there are larger multiproduct firms, offering several services ranging from data storage to the provision of standardised machine learning solutions. While prices do not appear to be the primary barrier to adoption, there are competition concerns relative to the potential switching costs associated with the bundling of these services (see, for example, Korinek and Vipra (2024^[25]) or André et al., (2025^[27])). As customers may face challenges in transitioning away from bundled offerings, this can lead to reduced competitive pressures and entrenchment of dominant firms within the market. Moreover, it is worth noticing that the largest players in this market (Microsoft Azure, Amazon Web Services, and Google Cloud Platform) accounted for about 65% of the market by mid-2023 (Biglaiser, Crémer and Mantovani, 2024^[52]).

Varian (2019^[47]) also provides a brief overview of the industrial organisation of firms providing AI services (akin to the concept of AI developers outlined above). A first issue is the possibility of customers lock-in. In Varian (2019^[47]), AI developers are mostly identified with cloud providers, which, besides data storage and provision, can offer standardised machine learning applications. The diversity of offered services might imply the presence of switching costs and, therefore, the ability for cloud vendors to raise prices. The issue does not seem particularly relevant for small and medium-sized applications, while it may be for larger and customised applications. The second aspect is relative to the pricing of cloud services. Even in the early phases, competition among cloud service providers has been relatively swift. The market dynamics might easily follow a Bertrand-type model for standardised services, with various providers trying to undercut the competitors.²⁵ At the same time, higher customisation might allow for a rise in the premia of the offered services.

Another perspective on the impact of AI development on competition, again mediated by productivity gains as in the case of AI use, is provided by the studies leveraging data on AI patenting by firms. Alderucci et al. (2020^[53]) find a positive impact of AI innovation on firm sales but a negative one on value added per worker. Damioli, Van Roy and Vertesy (2021^[54]) instead, find a positive impact of patent applications on labour productivity. The distinction between pure AI users and AI developers in these cases is somewhat blurred, as firms may develop AI tools primarily to support their own activities rather than as a core business, trying to solve the classical dilemma involving making or buying tools useful for their business. More direct evidence is provided by Calvino and Fontanelli (2024^[55]) who show that firms developing AI in France appear to realise productivity returns beyond selection, differently from firms buying AI from external providers. AI developers also appear to have a more specialised workforce (Fontanelli et al., (2024^[56])), which may contribute to explain why this group of firms already realises productivity returns, as well as may be a relevant determinant of within-industry concentration.

In the next sections, the analysis will try and relate some of the dimensions discussed in the conceptual framework proposed above to a range of data sources, further presented below.

Table 2.1. A proposed conceptual framework

		Users	Developers
Generative AI	Economic characteristics	Broad applicability (text drafting, image generation, coding); heterogeneous audience (ChatGPT vs. fine-tuning larger models). Gains from time saving and fast differentiated output, differences between use and productive use.	LLM development; compute, data, and skill requirements; high barriers to entry, economies of scales, strategic layers in the production chain (stack).
	Impact on competition/business dynamism	Potential disruption within certain industries (media, education). Performance boosting for incumbents, possibility for small players to enter the market.	Potential for concentration, oligopolistic dynamics, competition for resources, vertical integration.
Non-generative AI	Economic characteristics	AI use for efficiency, automation, decision; depends on domain data quality; improvements rather than radical novelty; embedded in enterprise core operations	Creators of domain-specific and task-oriented models. Relatively small ICT firms but also larger cloud platforms. Less compute-demanding than generative ones. Success based more on the integration with application rather than the model
	Impact on competition/business dynamism	Risk of dependency from AI providers. Potential role for data ownership and positive feedback loops. Adoption capability as key differentiator, leading to costs / productivity advantages and divides across competitors.	Domain expertise and proprietary data are key factors giving an edge to competitors.

3 Data

In order to carry out a first exploration of the links between AI and competition, the current analysis leverages a range of different data sources. This section discusses the key features of such data, as well as the relevant limitations, and presents the measures of competition used in the analysis.

First, the section focuses on ICT survey data and LEED. These data are used to obtain proxies for non-generative *use* and generative AI *exposure*, respectively. Then, it discusses patent data, used to proxy for AI-related patenting activity. Financial data of individual firms, used to compute firms' markups in a cross-country setting, are subsequently presented. Finally, this section introduces the market concentration measure, drawn from Calligaris et al. (2024^[57]). The section concludes with a methodological note on measuring competition.

Administrative data from France and Portugal

A key source of information to proxy for the use (or potential use) of AI by firms comes from administrative data, notably official ICT surveys and LEED.

Official ICT surveys are highly representative data containing relevant information on firms' use of advanced digital technologies, including AI. The common feature of these sources is their representativeness, as they are generally collected by National Statistical Offices, with the European countries following the same guidelines provided by Eurostat (Community Survey on ICT usage and e-commerce in enterprises). The data for these countries are thus generally based on common units of analysis and variable definitions and tend to overlap in terms of sectoral and firm size coverage.

The analysis presented in this paper relies on ICT surveys for two countries, France and Portugal, the *Enquête sur les Technologies de l'information et de la communication et commerce électronique* (TIC) and *Inquérito à Utilização de Tecnologias da Informação e da Comunicação nas Empresas* (IUTICE), respectively.²⁶ The surveys allow proxying AI use following responses to firms that are asked whether they used different types of AI technologies or not.²⁷

The TIC dataset for France provides stratified information on AI use for firms with at least 10 workers. It can be linked to firm-level administrative data, specifically the FARE balance sheet database, containing information for the population of French firms.²⁸ FARE contains yearly information on key financial variables, such as the firm's gross output, value added, employment and intermediate inputs. This paper uses the 2021 and 2023 TIC survey waves linked to the FARE dataset from 2011 to 2022.

Similarly, the IUTICE dataset for Portugal provides stratified information on AI use for all firms. It can be linked to the *Sistema de Contas Integradas das Empresas* (SCIE) dataset, a firm-level administrative financial data source, including balance sheet and other accounting data. These are collected every year for the population of firms in the private sector, except for the financial sector, from 2004 to 2023, covering around 400 000 firms per year. To ensure comparability with the French data, this analysis uses the 2021 and 2023 IUTICE data and the 2011-2022 SCIE data, and only considers firms with at least 10 workers. All monetary variables are deflated to 2005 USD values.

Since ICT surveys typically collect information that can be related to the previous calendar year, each wave is linked to the balance sheet data from that corresponding year. From these datasets, this paper draws a

binary indicator on whether each surveyed firm uses AI, as well as their respective markups (see Annex A for details on the estimation procedure) and market shares (see section 4 for methodological details).

Given the timing and characteristics of the above-mentioned data, it is not straightforward to untangle specifically generative AI use. This is why an additional data source is leveraged: LEED data. This source is available for Portugal from the Personnel Records (QP – *Quadros de Pessoal*). It is collected by the Portuguese Ministry of Labour, Solidarity and Social Security and provides information on workers in all Portuguese firms, except for the public administration, regional and local, as well as public institutions; and employers or workers of domestic service. The data include information on workers' formal education, age, gender, occupation, monthly wage, hours of work and the type of labour contract, from 1985 to 2023, with information about around 3 million workers per year.

Relevantly for the current analysis, a measure of *exposure* to generative AI is constructed to proxy for potential generative AI use by firms, based on their occupational composition. This is done following Felten, Raj and Seamans (2023^[58]), which provide scores of generative AI exposure at the occupational level. This methodology links specific human abilities (such as deductive reasoning or written expression) to the capabilities of LLMs, capturing the extent to which the tasks performed in a given occupation can be automated or augmented by generative AI.

Patent data

One way to identify firms engaged in AI development is through patent data. Although patents provide only a partial view of a firm's overall innovation activities, they remain a widely accepted proxy for innovative output (Basberg, 1987^[59]). Using data from PATSTAT Global – Autumn 2024 Edition, innovation is measured as annual flow, using the classification developed by the OECD STI Microdata Lab to proxy for AI-related innovations (OECD, 2025^[60]). A flow measure is preferred to a cumulative *stock* indicator, as the relatively short observation period limits the usefulness of depreciation-adjusted stock measures.

The number of AI-related innovations filed by each company is obtained by counting the annual number of DOCDB patent families each company files. A DOCDB family consists of patents filed in different patent offices that cover the same invention, and the filing date of the earliest patent in the family is used as the invention's date of inception. If multiple companies apply for the same DOCDB family, the earliest filing date is assigned to all co-applicants if they applied simultaneously. Otherwise, only the first applicant receives the earliest date, and subsequent applicants are assigned their respective application dates.

The analysis considers mostly patents linked to firms in the Moody's Orbis database. For limited descriptive evidence also the universe of AI-related application filings of kind A (applications made in national offices) are used, together with applications filed under the Patent Cooperation Treaty. AI-related patents are identified by combining the approaches developed in Baruffaldi et al. (2020^[61]) and by OECD (2025^[60]). They rely on a combination of AI keywords retrieved in the abstract or title of patent documents and on a selection of International Patent Classification (IPC) and Cooperative Patent Classification (CPC) codes related to AI. As mentioned above, it is important to recognise that patent data provide only a partial view of the innovative landscape. Not all inventions are protected through patents, and some innovations cannot be patented. In particular, in Europe, AI per se is often seen as a mathematical method and is excluded from patenting.²⁹ These limitations can be particularly relevant for AI-related inventions and may be due to the presence of open-source practices, different legal characteristics across jurisdictions, or the strategic choice to not disclose the content of the innovation.

Firm-level financial data

To compute firms' markups in a cross-country setting the dataset used is the Moody's Orbis database. It collects data on 450 million listed and unlisted firms worldwide, providing information on their economic activities, such as employment and output. The data used in the analysis are obtained from the 2023 Orbis vintage. Numerous steps are undertaken to clean the data, closely following Gal (2013^[62]), Kalemli-Özcan et al. (2024^[63]) and Calligaris et al. (2024^[10]).

Financial information within Orbis is available at the business group level (consolidated financial data aggregated across all firm subsidiaries) and at the individual firm-level (unconsolidated information referring to individual firms' activities). In this work, unconsolidated accounts are used to capture markups at the firm-level. For further analytical details on the estimation procedure, see Annex A.

To limit concerns over the representativeness of the dataset, the analysis includes only firms employing, on average, over the period at least five employees. This results in a final sample with information for 23 countries, Austria, Belgium, Bulgaria, Czechia, Denmark, Estonia, Finland, France, Germany, Hungary, Ireland, Italy, Latvia, Luxembourg, the Netherlands, Poland, Portugal, Romania, Slovenia, Slovakia, Spain, Sweden and the United Kingdom.

Ownership structure data

To compute measures of concentration in AI-related patenting, AI-related patents are aggregated across all subsidiaries within each business group, leveraging information on their ownership structures. Data on ownership structures come from the work by Calligaris et al. (2024^[10]). There, global ownership structures are reconstructed by identifying linkages between headquarters and subsidiaries. Orbis provides detailed ownership data linking each firm to its global ultimate owner, defined as the entity owning more than 50.01% of its shares. However, Orbis ownership information only covers the period 2007-2020. To get this information for the years pre-2007 and to further complement and quality-check the existing information, Orbis is supplemented with data from the Orbis M&A database, also provided by Moody's. This database has a global coverage starting from 2001, reports information on domestic and cross-border M&As and covers deals involving target and acquiring firms across all industries. Observing M&A activity between firms allows tracking changes in ownership over the years and the identification of business group structures pre-2007.³⁰

Start-up data

Data on start-ups are obtained from the OECD/STI Start-ups Database. This database combines data from Crunchbase and Dealroom, considered by investors and analysts among the most comprehensive sources of information on start-up activity in the US and Europe, respectively. Data are collected from various sources, such as public news, company filings, domain and trade registries, and investor portfolios, among others. Although Crunchbase and Dealroom were founded in 2005 and 2013, respectively, they contain retrospective information with a sufficiently large coverage from the early 2000s. These commercial datasets are combined, curated, cleaned, and harmonised to create a unique and comprehensive dataset on the worldwide start-up ecosystem. Extensive information on the database, its coverage, and its content is available in Berger et al. (Forthcoming^[64]).

To ensure the analysis captures innovation- and growth-oriented start-ups – following an extensive literature (Fons-Rosen, Roldan-Blanco and Schmitz, 2021^[65]) – the OECD/STI Start-ups Database is further refined selecting only venture capital-backed (henceforth, VC) or patenting entities. The time horizon, depending on the different exercises, captures start-ups that have received the first VC round or

filed its first patent between 2001 and 2024. In total, this paper considers 382 108 start-ups fulfilling these criteria.

To characterise the dynamism of the entrepreneurial landscape, key variables used in the paper include the number of entities founded across years and the amount of VC funding received. The former is a good indicator of potential market entry, while the latter is a proxy for the potential disruption and quality of new entities. Finally, the analysis also examines acquisition patterns. Acquisition information is obtained from the OECD Start-up database, complemented with Orbis M&A, and includes all transactions in which at least 50.01% of start-up shares are purchased by a single acquiror.

Start-ups in the sample are categorised as GenAI, non-GenAI, or nonAI using information on tags, business descriptions, and patent data. Crunchbase and Dealroom provide qualitative tags – mostly single words – describing each firm’s main activities or technologies. The tags “AI”, “Artificial Intelligence”, “Gen AI”, and “Generative Artificial Intelligence” are used to identify AI-related entities. In addition, the OECD Start-up database provides patent information: each patent is classified as non-GenAI following the procedure described above, or as GenAI if it matches the list of patents identified by the World Intellectual Property Organisation as Generative AI-related.

Building on these sources, a machine learning approach is applied, based on a large language model (LLM) to classify start-ups by AI type. The LLM is pre-trained on business descriptions of all entities in the OECD Start-up database, allowing it to learn how to read and interpret domain-specific language. It is then fine-tuned on a labelled subset of start-ups using the tag- and patent-based classifications described above. Through this process, the model learns the textual and contextual patterns that characterise GenAI, non-GenAI, and nonAI entities. This enables it to detect AI-related activities even when these are not explicitly referenced in the tags, thus mitigating misclassification due to sparse or noisy tagging.

At the end of the procedure, 5 633 start-ups are identified as GenAI (1.5% of the sample), 89 764 as non-GenAI (23.5%), and the remaining 75% as nonAI start-ups.

Other data

Market concentration data

Data on market concentration are obtained from Calligaris et al. (2024_[66]) and (2024_[57]), who innovate with respect to the previous literature by using more narrowly defined industries, going from 2-digit to mostly 3-digit levels. In addition, such data allow accounting for the international dimension of competition when defining markets, relying on a taxonomy of industries that defines whether each industry competes mainly at the domestic, European, or global level. The higher disaggregation at the industry level and the identification of the geographical boundaries of each market allow to approximate the so-called relevant market more precisely.³¹ Countries included in the sample are Belgium, Denmark, Finland, France, Germany, Greece, Hungary, Italy, Norway, Poland, Portugal, Slovenia, Spain, Slovenia, Sweden, and the United Kingdom for the domestic and European buckets, while in the global one also Japan, Korea, and the United States are included. For industries classified as competing at the European or global level, there is only one market (and, thus, one observation) per year, whereas for industries competing at the domestic level, concentration is measured for each country and year separately. The period covered is 2000-19.

Box 3.1. Measuring competition: a methodological note

Competition is a complex concept, and no single variable can fully capture it. The literature has adopted several proxies, such as industry level concentration ratios, firms' market share, and markups, each reflecting different aspects of competitive dynamics. This report uses several indicators, chosen according to the nature of the specific exercise and data limitations. In particular, a distinction must be made when using concentration measures as, despite their conceptual simplicity, their careful measurement requires several assumptions, which are not always met due to the data limitations.

These challenges are particularly relevant for the analysis on AI use and competition of section 4, which uses individual firms' market shares. First, due to data limitations, the geographical extent of the market is not accounted for, although the lack of a cross-country dimension in this part of the analysis makes it less concerning. While openness to trade was not explicitly accounted for in the different empirical exercises, the use of sector and country fixed effects should mitigate such a concern. Second, unlike other recent studies of trends in concentration (Bajgar et al., 2023^[11]; Calligaris et al., 2024^[10]), the business group dimension is not taken into account due to the nature of the data. Such a limitation may result in an underestimation of market shares and concentration. Moreover, the use of market shares as proxies of competition must be considered with caution, as they just capture the relative importance of single firms in terms of sales within a market. However, the analysis looking at the link between AI adoption and market share can be considered as a first step to understand the impact of AI adoption on concentration. In fact, given that AI adoption is likely to happen among firms with high market share, which are those driving concentration measures, this analysis could relevantly provide first indirect evidence on the link between AI and concentration. These concerns do not apply to the exercises on AI development and competition, where more aggregate concentration measures are instead taken from Calligaris et al. (2024^[57]), and account for both the geographic and the business groups dimensions. Finally, the competition proxies may incorporate a short-term cyclical component in their time variation. The inclusion of year fixed effects across all empirical strategies should help account for this.

A word of caution should be added about the ability of concentration and market shares to proxy for competition. While largely used in policy analysis and academia (OECD, 2021^[67]), the theoretical relationship between competition and concentration and market shares is not univocal. Considering textbook models of competition between firms, high concentration can be associated with competitive outcomes (as in the Bertrand model) or with lower competition (as proxied by a lower number of firms, which raise markups and equilibrium market shares, in Cournot model). For an early discussion on the topic, see Demsetz (1973^[68]), who pointed out how high concentration can be the outcome of a competitive process, with firms gaining market shares outperforming rivals, and Schmalensee (1989^[69]), who reviewed the early literature relating concentration with other proxies of competition. For recent assessments, see Berry et al. (2019^[70]) and Syverson (2019^[71]), who provide an overview of the advantages and drawbacks of using concentration to assess market power.

4 Evidence from microdata

Building upon the conceptual framework discussed in Section 2, this section aims at providing a first picture of the relationship between AI and competition, leveraging the different sources of microdata described in Section 3. Given data availability, the analysis is not able to comprehensively cover all dimensions of the conceptual framework. In particular, the data at hand do not allow to zoom in on competition dynamics among foundation models – which are object of complementary analysis (André et al., 2025^[27]). Also, the focus is on AI itself rather than on the broader AI value chain (OECD, 2025^[23]). Despite these caveats, leveraging information from different sources of microdata across countries – notably ICT surveys on AI use, LEED data on occupational exposure to generative AI, patent data identifying AI-related innovations, as well as AI start-up data – allows providing a uniquely broad overview of different facets of the links between AI and competition in the age of AI.

Evidence on AI use from official ICT surveys

As a starting point, the analysis leverages information on AI use from firm-level official ICT surveys. AI use is therein proxied by a binary indicator indicating whether firms use AI or not, based on detailed questions that follow Eurostat's guidelines for AI use measurement.³² Given the time horizon considered in the available surveys (2020 and 2022), this exercise mainly captures the use of non-GenAI.³³ ICT surveys are then matched with balance sheet data, focusing on France and Portugal, countries for which both sources are available and accessible. This allows focusing on the links between AI use and two proxies of competition, markups and market shares, at the firm-level.

In terms of measurement, market shares are computed as the ratio between firms' and industry's sales at the 3-digit level.³⁴ To account for high heterogeneity in market size across sectors, relative ranks are used. This approach is relevant from a competition perspective, as changes in market shares may have different competition implications depending on firms' initial levels. Specifically, each firm is positioned into a market share decile within its respective sector. This method categorises firms into 10 groups, from the bottom 10% (the first decile) to the top 10% (the tenth decile). Firm-level markups, which theoretically measure the wedge between a firm's output price and its marginal cost and provide a proxy for market power, are instead estimated using the methodology proposed by De Loecker and Warzynski (2012^[72]) – see Annex A for details on the estimation procedure.

To explore the links between AI use and the two competition proxies considered, the following equation is estimated separately for each country:

$$Y_{i,t} = \beta_0 + \beta_1 AI_{i,t} + \beta_2 X_{it} + \zeta_j + \gamma_k + \varphi_s + \delta_t + \varepsilon_{i,t}$$

Equation 1

For each firm i in year t in the sample. $Y_{i,t}$ alternatively represents market share decile levels $s_{i,t}$ and dynamic changes of market shares and markups. In the dynamic specification, $Y_{i,t}$ captures the evolution of market shares or markups over a five-period window, starting three years prior to the moment in which AI use is observed and extending to two years after.³⁵ Specifically, when analysing market share dynamics, the dependent variable assumes the value of -1 if the firm's decile level decreased, 0 if it remained the

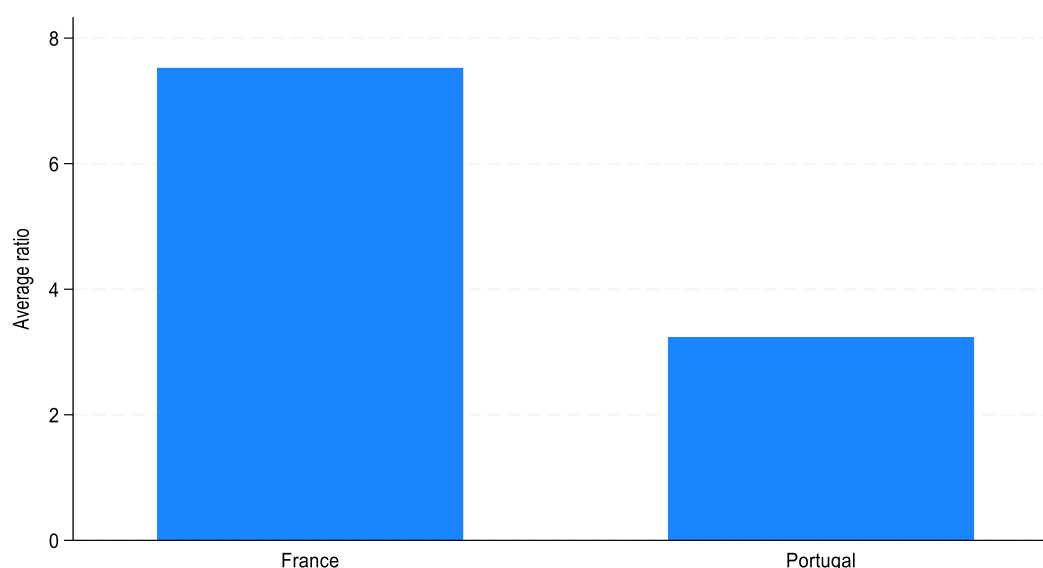
same, and 1 if it increased. The time window's upper bound ($t + 2$) is determined by data availability, as the earliest year in which AI use is observed in both countries is 2020 and balance sheet data at the time of writing are available until 2022. The lower bound ($t - 3$) is chosen to capture firm performance over a longer period, also considering that the exact timing of AI adoption is unknown.

The key independent variable is $AI_{i,t}$, a dummy variable that indicates whether a firm uses AI or not.^{36 37} X_{it} accounts for the use of fast broadband (broadband connection with at least 100 Mbit/s) to capture digital infrastructure and the share of other digital technologies adopted by each firm. This share is computed as the number of digital technologies that firm i uses in year t out of those surveyed (3D printing, big data analysis, cloud computing services, internet of things, robotics, customer relationship management, enterprise resource planning, and e-commerce). Additionally, when focusing on dynamic changes in market share deciles, the decile level in ($t - 3$) is also included as a control, as firms in higher deciles have less margin for relative improvement. The model also includes firm size and firm age, sector (φ_s , at the A38 level of aggregation), and year (δ_t) fixed effects. Robust standard errors are employed to account for heteroskedasticity.

Market shares

Firms using AI tend to have higher market shares than other firms. This is evident from Figure 4.1, which plots the ratio of the average market share of AI users relative to that of non-users in France and Portugal. The data indicate that firms using AI have market shares that are 7.5 and 3.2 times larger than non-users in France and Portugal, respectively. This striking disparity motivates the need to investigate whether AI users maintain this competitive edge even after controlling for firm characteristics, namely size and age. Furthermore, it prompts further analysis of whether distinct trends in market share growth rates can be observed between these two groups of firms.

Figure 4.1. Ratio of the average market share of AI users to non-users: France and Portugal



Note: The chart shows the ratio of the average market share of AI users to non-users of AI. Market share is proxied by the ratio of a firm's sales to its 3-digit industry's total output. Only firms with at least 10 workers are considered.

Source: Authors' calculations using data from *Enquête TIC* and *FARE* for France, and *ITUCIE* and *SCIE* for Portugal

Table 4.1 extends therefore the descriptive analysis above by further focusing on firms' market shares and estimating Equation 1 using market share decile as dependent variable. The baseline regression estimates, obtained using an Ordinary Least Squares (OLS) model, in columns 1 and 4, show that AI adopters are on average associated with a higher position in the market share distribution.³⁸ However, when focusing on the relative position of firms in the market share distribution (see Table 4.1), this premium tends to lose statistical significance once additional controls are included (columns 3 for France, and 6 for Portugal). Such controls include the use of fast broadband, the firm's digitalisation intensity, and its lagged productivity. This suggests that the positive association found in the baseline estimates might be due – at least to some extent – to selection dynamics, as AI-adopting firms tend to be already more digitalised and productive. These results resonate with the cross-country analysis by Calvino and Fontanelli (2023_[4]), which suggest that AI adopters are more productive than other firms, but such productivity premia are not entirely credited to AI use.

Using an ordered probit model to further investigate heterogeneity among firms with different positions in the market share distribution provides a more nuanced picture. After accounting for the full set of control variables as in columns 3 and 6, estimates suggest that using AI is associated with a higher likelihood of being in the top decile, in both France and Portugal. However, this relationship is only statistically significant for France. This suggests that, although selection dynamics might be relevant, the impact of AI may be heterogeneous and stronger for firms at the top of the market share distribution.

Table 4.2 shifts the focus to market share dynamics, using changes in firms' position in the market share distribution as dependent variable. Focusing on the main coefficient of interest, *AI users*, the findings suggest that adopting firms are generally no more likely than non-adopters to move up in the market share distribution. Therefore, AI adoption – on average – does not appear related to substantial increases in terms of market share ranks within the five-year period considered.

Focusing on other control variables, the results in Table 4.2 also suggest that firms with higher initial market shares are less likely to improve their relative position. This is shown by a consistently negative and significant coefficient on the initial decile level, which was included in the models to account for this mechanical ceiling effect. Being more productive is also associated with increases in firms' position in terms of market shares, as highlighted by the positive and significant coefficient for lagged productivity. Robustness checks using an Ordered Probit model qualitatively confirm the above-mentioned findings.³⁹

Table 4.1. Firm's market share decile and AI use: France and Portugal

	France				Portugal	
	(1)	(2)	(3)	(4)	(5)	(6)
AI user	0.0657*** (0.0204)	0.0530** (0.0208)	0.0266 (0.0178)	0.0681*** (0.0208)	0.0283 (0.0214)	0.0199 (0.0206)
Fast broadband		0.0122 (0.0160)	-0.0029 (0.0148)		0.0459* (0.0251)	0.0324 (0.0242)
Tech intensity		0.0769*** (0.0213)	0.0596*** (0.0193)		0.2251*** (0.0362)	0.1068*** (0.0357)
L3.Log(Productivity)			0.1816*** (0.0134)			0.1925*** (0.0140)
N	15 837	15 837	14 895	8 472	8 472	8 144
Adj. R ²	0.34	0.34	0.34	0.33	0.34	0.37

Note: Robust standard errors in parentheses, *** p<0.01, ** p<0.05, * p<0.1. The dependent variable is the market share decile within a 3-digit sector. Fast broadband stands for the use of a broadband connection with at least 100 Mbit/s. Tech intensity stands for the share of other digital technologies adopted. Although not reported, all models include controls for firm size (in terms of employment), firm age, as well as industry and year fixed effects. The analysis leverages ICT firm-level information for 2020 and 2022 for both countries.

Source: Author's calculations using data from Enquête TIC, FICUS/FARE, ITUCE and SCIE.

Table 4.2. Changes in firm's position in the market share distribution and AI use: France and Portugal

	France			Portugal		
	(1)	(2)	(3)	(5)	(5)	(6)
AI user	0.0124 (0.0094)	0.0091 (0.0100)	-0.0052 (0.0101)	0.0112 (0.0151)	0.0039 (0.0153)	0.0045 (0.0150)
L3. Decile	-0.1185*** (0.0054)	-0.1186*** (0.0055)	-0.1367*** (0.0074)	-0.1494*** (0.0098)	-0.1506*** (0.0098)	-0.1714*** (0.0121)
Fast broadband		0.0013 (0.0091)	-0.0011 (0.0091)		0.0153 (0.0159)	0.0147 (0.0159)
Tech intensity		0.0183 (0.0171)	0.0189 (0.0173)		0.0428* (0.0259)	0.0271 (0.0262)
L3.Log(Productivity)			0.0074*** (0.0058)			0.0230** (0.0083)
N	7 550	7 550	7 333	3 659	3 659	3 591
Adj. R ²	0.31	0.31	0.12	0.15	0.15	0.14

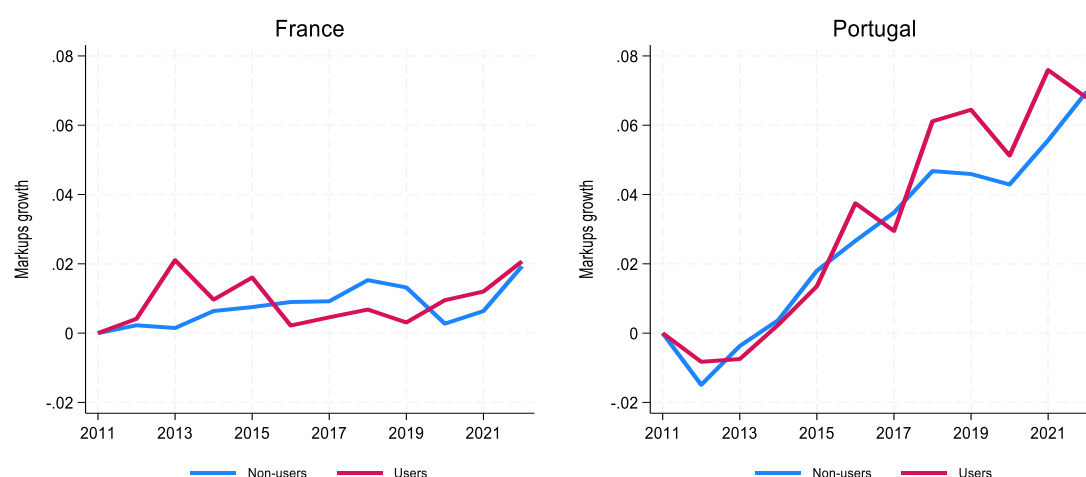
Note: Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is the five-year change in market share decile level within a 3-digit sector. Fast broadband stands for the use of a broadband connection with at least 100 Mbit/s. Tech intensity stands for the share of other digital technologies adopted. Although not reported, all models include controls for firm size (in terms of employment), firm age, as well as industry and year fixed effects. The analysis leverages ICT firm-level information for 2020 for both countries.

Source: Author's calculations using data from Enquête TIC, FICUS/FARE, ITUCE and SCIE.

Markups

Turning the attention to markups, Figure 4.2 presents descriptive evidence on their cumulative growth rates from 2011 to 2022. The figure compares the trajectories for two relevant groups of firms, defined by their AI use status in 2020: AI adopters and non-adopters. Consistent with Calligaris et al. (2024^[10]), the overall markup trends are quite different between the two countries, increasing substantially more in Portugal than in France. Furthermore, within each country, the markup paths of AI adopters and non-adopters have not significantly diverged over the period.

Figure 4.2. Markups growth rate for Users and Non-users of AI



Note: The chart shows log changes of unweighted average markup across firms that did not adopt AI and firms that adopted AI. The figures plot log markups and the 2011 level is normalised to 0. Hence, the vertical axes represent log-differences from the starting year and approximate percentage growth rates. Only firms with at least 10 workers are considered.

Source: Authors' calculations using data from Enquête TIC and FARE for France, and ITUCIE and SCIE for Portugal.

The lack of association between AI and markup growth is confirmed by the econometric results in Table 4.3, which go beyond the descriptive analysis presented in the Figure. Estimating Equation 1 using as dependent variable the growth rate of markups between $(t - 3)$ and $(t + 2)$ shows no statistically significant relationship between AI adoption and markup growth in either country.⁴⁰

Table 4.3. Growth in firm's markup and AI use: France and Portugal

	France		Portugal	
	(1)	(2)	(3)	(4)
AI user	0.0081 (0.0064)	0.0073 (0.0065)	0.0163 (0.0126)	0.0076 (0.0131)
Fast broadband		0.0035 (0.0041)		0.0156 (0.0105)
Tech intensity		0.0030 (0.0083)		0.0529*** (0.0185)
N	6 020	6 020	2 828	2 828
Adj. R ²	0.03	0.03	0.01	0.01

Note: Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is the growth rate of markups between $t+2$ and $t-3$. Fast broadband stands for the use of a broadband connection with at least 100 Mbit/s. Tech intensity stands for the share of other digital technologies adopted. Although not reported, all models include controls for firm size (in terms of employment), firm age, as well as industry and year fixed effects. The analysis leverages ICT firm-level information for 2020 for both countries.

Source: Author's calculations using data from Enquête TIC, FICUS/FARE, ITUCE and SCIE.

While the results presented above on the links between AI, market shares and markups should be interpreted with caution, due to the time period under scrutiny and the lack of a causal identification strategy, they provide suggestive evidence that – on average and in the period under analysis – AI use by firms does not seem related to substantial increases in proxies of market power in France and Portugal.⁴¹ This is particularly true for markups trends, which do not appear significantly different between AI users and other firms, while for market shares dynamics some specifications point to a possible role of AI strengthening the edge of most competitive firms. This reinforces the relevance of continuing monitoring such dynamics in the future. As previously mentioned, the binary AI use proxy employed in the above estimates is not able to capture the recent boom in generative AI use, which will be further discussed in the next subsection.

Evidence on occupational exposure to GenAI from LEED

Official statistics on GenAI use by firms are not yet widely available, although a rapid uptake of such tools has been recently observed.⁴² Among the different strategies that researchers have developed to study the evolution of GenAI, this paper builds on recent literature looking at the impact across workers' occupations more affected by GenAI. Specifically, this exercise leverages recent LEED for Portugal, linked with the previously described official ICT surveys and balance sheet data, to characterise the firms most exposed to GenAI along several dimensions. This characterisation is essential to assess potential competition implications: if high exposure overlaps with existing dominance (e.g. high market shares), GenAI could entrench incumbents' positions; conversely, if it primarily affects smaller and young firms, it may foster competition.

Firm-specific GenAI exposure is measured by firms' occupational composition, following Felten, Raj and Seamans (2023^[58]), who provide a score for each occupation of how likely it is to be exposed to GenAI. This measure is agnostic with respect to whether exposure is more likely to result in automation or augmentation of occupations.⁴³ In particular, firm-level exposure is defined as:

$$\Phi_j = \sum_p \omega_p^j \times \phi_p$$

Where Φ_j represents the GenAI exposure level of firm j , ϕ_p represents the exposure index score of occupation p , ω_p^j represents the occupation weight (in terms of number of employees) of occupation p within firm j and $\sum_p$ represents the summation across all occupations p present within firm j .

Exploring the characteristics of firms with a different degree of exposure to generative AI is a first step to better understand the links between generative AI use and competition. In this context, Table 4.4 presents information on several firm characteristics – notably including firm turnover, value added, capital stock (measured by fixed assets), employment, labour productivity (measured by the ratio of value added per worker), firm age, firms' market shares within 3-digit sectors, as well as the shares of firms that report to use AI and their share of tertiary educated workforce. To ease interpretation, this information is reported by quintiles of GenAI exposure, with the first quintile representing firms with the lowest level of exposure and the fifth quintile representing firms with the highest level.

Table 4.4. Firm's characteristics by generative AI quintile - Portugal

Generative AI quintile	Q1	Q2	Q3	Q4	Q5
AI users	0.06	0.09	0.08	0.09	0.12
Turnover (EUR ths)	961	1 851	1 651	1 614	1 077
Value added (EUR ths)	376	597	661	650	497
Capital (EUR ths)	336	666	595	480	367
Employment	14	16	15	9	6
Productivity (EUR ths)	24	32	37	45	52
Firm age	15	17	18	16	12
Market share	0.06	0.11	0.09	0.09	0.05
Markup	1.58	1.57	1.64	1.87	1.95
Tertiary education share	0.04	0.09	0.16	0.30	0.46

Note: The analysis leverages information for 2023. Productivity is defined as the ratio of valued added per worker.

Source: Author's calculations using data from ITUCE, SCIE and QP.

Firms more exposed to GenAI are more likely to use AI, as measured by the firm-level ICT survey information used in the previous subsection.⁴⁴ The share of AI users appears in fact twice as high in firms with highest exposure to generative AI with respect to those with lowest exposure. This may on the one hand reflect the fact that technological relevance is a driver of adoption, and on the other hand reflect sectoral composition, with highly exposed occupations being prevalent in information services sectors, which tend to have high AI use.

Firms more exposed to GenAI are also on average smaller in terms of employment than the least exposed ones, as well as to some extent younger – although this is only evident for the highest exposure quintile. Although descriptive, this evidence suggests that generative AI may represent a relevant opportunity for smaller and younger firms.

Furthermore, firms more exposed to GenAI appear more productive and have a higher share of their workforce with tertiary education. This strong correlation is largely driven by the nature of the exposure measure itself: unlike previous waves of automation that often-targeted routine tasks, GenAI capabilities can overlap more significantly with cognitive, non-routine tasks typically performed by highly skilled workers (e.g. programming, drafting, analysis). Consequently, firms that employ such professionals naturally register higher exposure scores.

This observation carries significant implications for market dynamics. It suggests that the benefits of generative AI may be unlocked more easily by firms that have higher capabilities, thereby raising the risk of widening productivity gaps and increasing concentration at the top of the distribution. However, with respect to turnover, value added, capital stock, and market shares, the evidence appears more mixed as

an inverted U-shaped pattern is observed, with firms in the middle of the exposure quintiles typically presenting highest values. This suggests that high exposure is not yet exclusively the domain of the largest incumbents. From a competition standpoint, this could indicate a window of contestability where mid-sized, productive firms leverage these tools to challenge market leaders.

This initial descriptive evidence should be considered as exploratory, and further OECD work relating firm characteristics with generative AI exposure is ongoing, also broadening the scope of analysis beyond Portugal.

Evidence on AI innovation from patent data

This section describes the relationship between firms' patenting activity – with a focus on AI-related patent filings, also linked with firms' financials – and various proxies of competition. While patent data are a widely used proxy of innovative activities, several caveats need to be taken into account when interpreting the results. First, there are inherent lags in the reporting of both patents and firms' financials. This implies poorer coverage in the latest years of the sample, when AI became more widespread (in particular concerning LLMs and generative AI), limiting the ability to fully capture the impact of AI innovation on competition. Moreover, even for inventions developed years before the end of the sample, effects on the competition proxies might still have to fully materialise. Second, AI-related inventions are concentrated in a relatively limited number of sectors. This limits data availability, especially when considering market concentration measures which are defined at the industry level.⁴⁵

Patenting and industry concentration

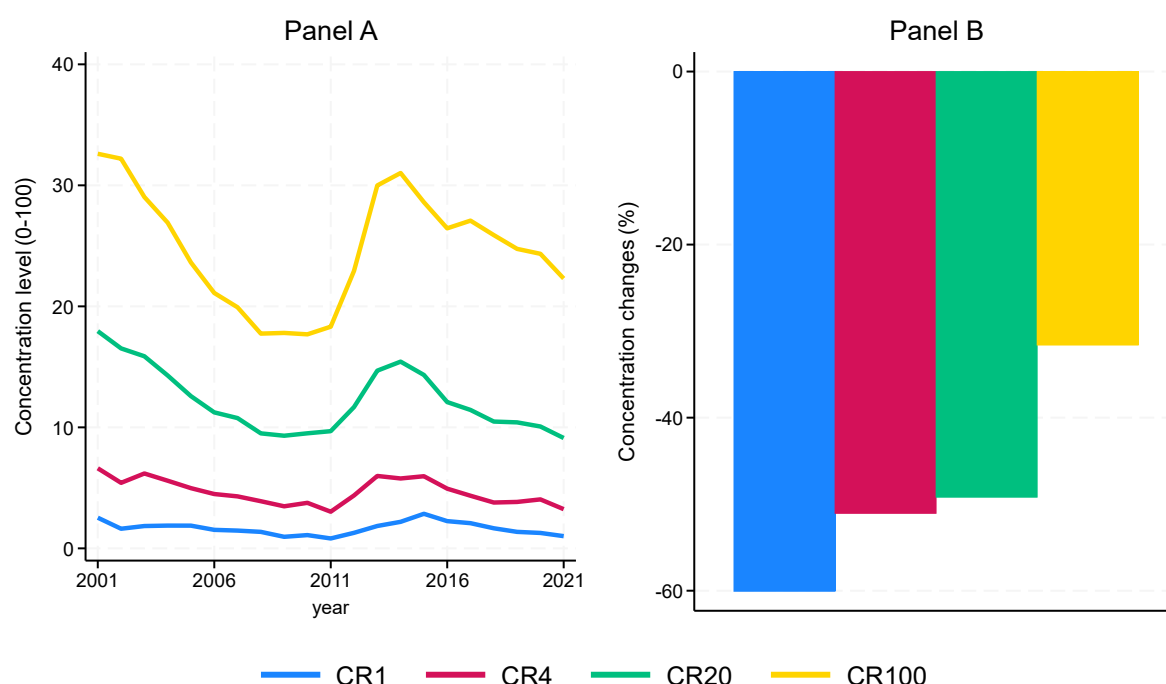
A first indicator of the state of competition among AI innovators is given by a global index of patent concentration. This index captures the share of AI-related innovations owned by the largest entities, based on the annual flow of patent applications. Specifically, it is calculated as the share of patents associated with the largest patentees relative to the total number of patents. Figure 4.3 reports this index and its evolution over time for different groups of patentees: the largest single one (CR1), the top 4 (CR4), the top 20 (CR20), and the top 100 (CR100).

Panel A of Figure 4.3 reports the trends of these measures over the years 2001-21. The early years of the sample display relatively high concentration, likely reflecting the presence of only a limited number of AI innovators. The various indicators decline until 2011, suggesting that an increasing number of firms became involved in AI innovation activities. After 2011, often considered a turning point of AI development (Engberg et al., 2024^[73]), concentration sharply increases according to all indicators, possibly reflecting substantial innovations efforts in this rapidly growing technology domain. Notably, the increase in CR100 is the largest among the concentration measures, suggesting that the rise in concentration was not driven solely by a few dominant AI innovators, but was instead spread among the top 100 patentees. Nevertheless, focusing on the latest years of the sample, a decline in concentration level across all indicators can be observed.

This interpretation is consistent with the evidence reported in Panel B of Figure 4.3. First, it shows that the percentage change in concentration between 2001 and 2021 is negative for all indicators and that the 2021 levels, despite the rapid increase after 2011, are still within the historical ranges. Second, this long run comparison shows that concentration declined relatively less for the broader categories (e.g. CR4 declined relatively less than CR1, but more than CR20). This pattern suggests that, over the sample period, not only an increasing number of AI-related patents have been filed by relatively smaller actors (as reflected in declining concentration measures), but also that medium-sized players have gained relative importance, leading to a less skewed distribution of patents filing among largest patentees. This pattern appears to be consistent with the characteristics of an expanding market.

These figures are based on the universe of the AI-related patent filings from PATSTAT. For robustness, Figure A B.1 of Annex B reports analogous graphs based on AI-related patent applications filed under the Patent Cooperation Treaty (PCT), which guarantees better cross-country comparability.⁴⁶ As in Figure 4.3, the trends for each indicator initially decline and then rise sharply after in 2011 (panel A of Figure A B.1). One difference between the two figures is that concentration level based on PCT data remains relatively high in the years following 2011.

Figure 4.3. Trends of concentration in AI patenting activity



Note: In Panel A, the lines represent the AI-related patent concentration ratio, calculated as the share of patents associated respectively, the largest single one (CR1, blue), the top 4 (CR4, red), the top 20 (CR20, green), and the top 100 (CR100, yellow). In Panel B, the bars represent the growth rate, between 2001 and 2021, of each indicator of Panel A. The graphs are based on Patents of Invention applications of kind A (national applications), contained in PATSTAT, and associated with the AI tag developed by OECD (2025_[60]).

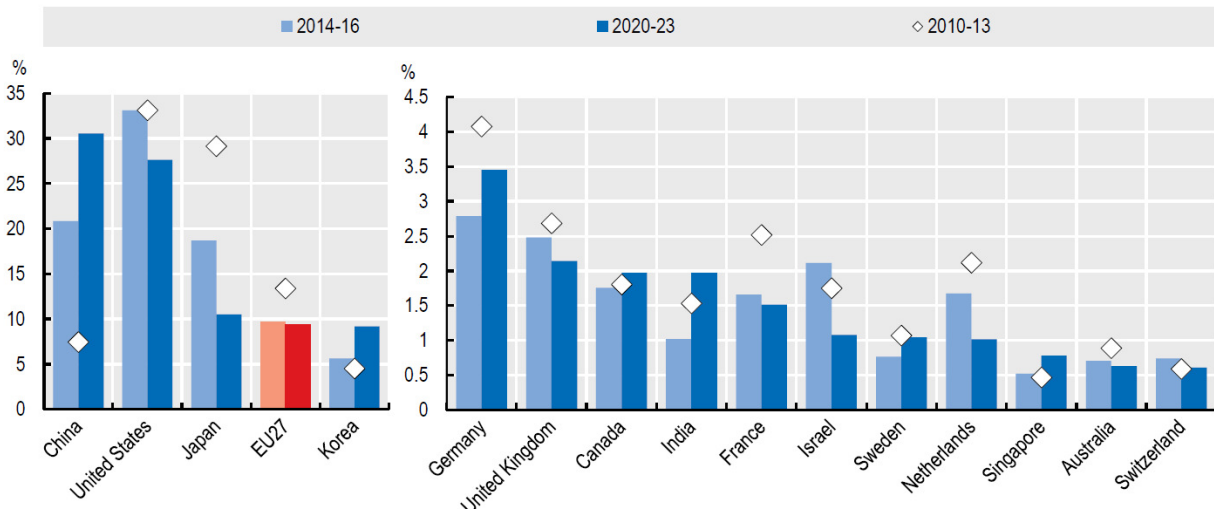
Source: OECD calculations based on PATSTAT Global.

Nevertheless, looking at the long run growth rate for the different indicators (Panel B of Figure A B.1), changes remain broadly proportional across measures, with CR1 declining over 2001-2021, and the other indicators increasing. Overall, the message seems complementary to the one obtained from Figure 4.3: the increase in concentration is stronger in the broader indicators, suggesting that medium-large patentees (i.e. those ranked between 21st and 100th position) have experienced a relatively larger increase in their patenting shares. On the other hand, the long run increase in concentration across most of these indicators calls for future monitoring, as it might lead to higher licensing and legal burdens through practices such as patent thickets and, in turn, to harder entry for start-ups and, more generally, to declining innovation.

Figure 4.4, sourced from OECD (2025_[60]), takes a different perspective and shows the evolution of the AI innovation share across countries (and regions) over the years. The figure shows that AI-related inventions are nowadays concentrated mostly in the United States and in China, which alone account for almost 70% of the filings in the latest years of the sample. While China's share increased markedly during the last decade, the role of the United States and Japan dropped. EU27 maintains a relatively low and constant

share of AI-related patent filings. Such an increasing concentration in a limited number of countries might suggest a widening gap in AI innovation, with potential increased dependency of the lagging countries from the leading one, with implications for productivity and competitiveness (Draghi, 2024^[74]).

Figure 4.4. Country share of AI inventions



Note: Data refer to patent applications filed under the Patent Cooperation Treaty (PCT), by priority date and inventor's location, using fractional counts. Figures for 2020-23 are estimated.

Source: OECD (2025^[60]).

While the previous section presents mixed findings, emphasising the need for future monitoring of concentration in AI innovation, concentration of AI patenting activity in a small number of large firms remains a phenomenon that is worth monitoring, especially considering its potential ties with market concentration. AI innovation appears to be positively related to firm size (Babina et al., 2024^[43]), due to a stronger ability by large firms to invest and to the presence of complementary assets and infrastructure. The advantages brought in by AI innovation might therefore provide a disproportionate edge to firms that already dominate their market. The following section relates two measures of patenting concentration to a measure of market concentration, adopted from Calligaris et al. (2024^[57]), and defined as:

$$CR_{s,g,t}^4 = \frac{\sum_{f \in Top\ 4} S_{f,s,g,t}}{S_{s,g,t}}, \quad \text{Equation 2}$$

Where the denominator $S_{s,g,t}$, is defined as the total gross output in an industry s , in its relevant geographic market g , at time t , and the numerator is the sum of sales of top four business group f operating in the market. Here the “market” identifies the combination of industry s and geography g (domestic, European or global) in which firms operate.⁴⁷

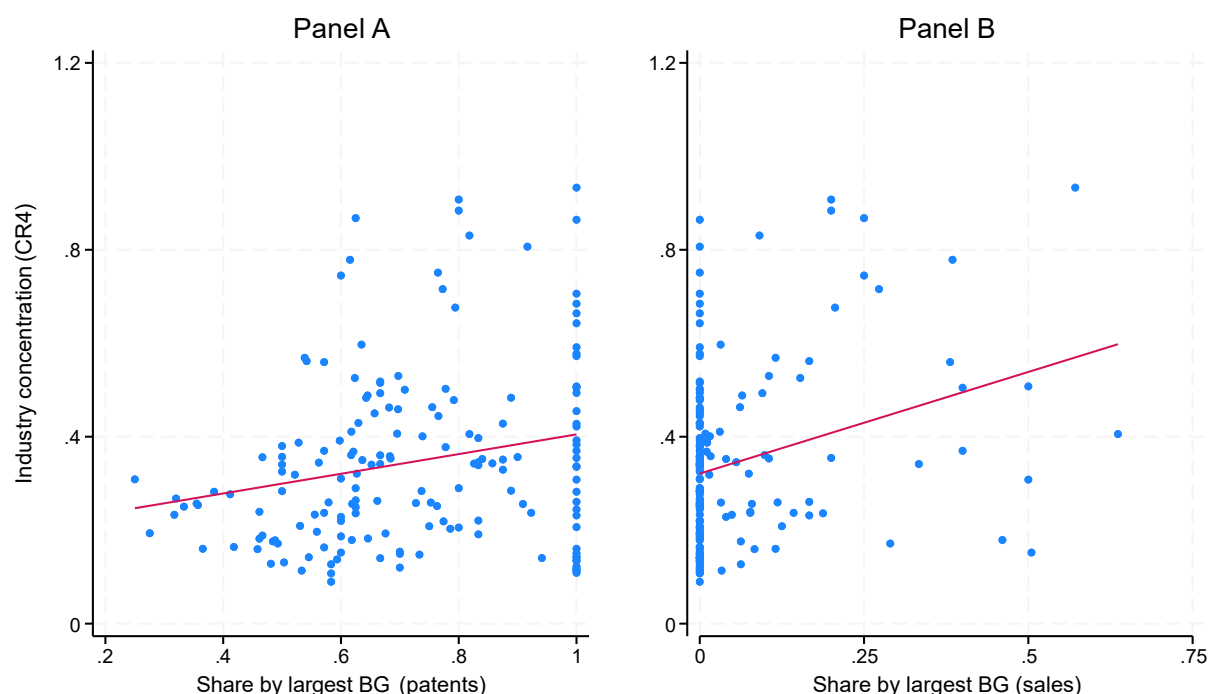
Figure 4.5 and Table 4.5 explore the relationship between market concentration and two different measures of AI patenting concentration.⁴⁸ Table 4.5 reports the estimates of the following equation:

$$CR_{s,g,t}^4 = \beta_0 + \beta_1 PAT_{s,g,t}^k + \tau_t + \gamma_g + \varepsilon_{s,g,t}, \quad \text{Equation 3}$$

Where $CR4_{s,g,t}$ is market concentration in industry s , geographical region g , and year t , computed following Calligaris et al., (2024^[66]). τ_t and γ_g are year and geographical region fixed effects.⁴⁹ Finally, $PAT_{s,g,t}^k$, with $k \in \{PL, ML\}$, represents two measures of AI-related patenting activity concentration computed at the same level of market aggregation as the concentration measure. $PAT_{s,g,t}^{PL}$ represents the share of AI-related inventions associated to the largest business groups in terms of patents (patent leaders, in short), and it's the market level analogue of the global indicator represented in Figure 4.3. $PAT_{s,g,t}^{ML}$ instead, is the share of AI-related inventions owned by the largest business groups in terms of sales (patents to market leaders in short).

Panel A of Figure 4.5 shows a positive correlation between the share of AI-related inventions owned by the largest business groups in terms of patents ($PAT_{s,g,t}^{PL}$, on the x-axis) and market concentration (y-axis). The first variable can be thought of as a measure of concentration in innovation, as it measures the extent to which AI innovation is spread among different firms in an industry. The relationship is also analysed through the estimates of Equation 3 in Panel A of Table 4.5. The regression confirms the positive relationship between the two variables, which is robust to the inclusion of year and geographical market fixed effects. Panel B of Figure 4.5 displays a positive relationship – also reflected in Panel B of Table 4.5 – between the share of AI-related inventions associated to the largest business groups in terms of sales ($PAT_{s,g,t}^{ML}$, on the x-axis) and market concentration (y-axis), and might suggest a link between concentration and AI innovation.

Figure 4.5. Market and AI-related patenting concentration



Note: The graph in Panel A represents the relationship between the share of AI-related inventions associated to the largest business groups in terms of patents (patent leaders in short, x-axis) and market concentration (y-axis). The one in panel B represents the relationship between share of AI-related inventions owned by the largest business groups in terms of sales (patents to market leader in short, x-axis) and market concentration (y-axis). This sample includes 15 industries (mostly from Manufacture of basic pharmaceutical products; Manufacture of computer, electronic and optical products; Professional, scientific, and technical activities) over the years 2000-2019. Each industry is classified either as domestic, European, or global, according to the taxonomy by Calligaris et al. (2024^[66]). The countries included in the sample are Denmark, Finland, France, Germany, Greece, Hungary, Italy, Norway, Poland, Portugal, Slovenia, Spain, Sweden, and the United Kingdom for the domestic and European buckets, while in the global one also Japan, Korea, and the United States are included.

Source: OECD calculations based on PATSTAT Global and Moody's Orbis.

Even though market leaders do not seem more likely to develop AI innovation (see the blue histogram in Figure A B.1, showing that in the majority of industries few or none of the AI-related patents are owned by market leaders), this phenomenon becomes more frequent in concentrated industries.

This result might have several interpretations. On the one hand, market concentration might be reinforced by the advantages brought in by AI patenting by market leader. On the other hand, leader of concentrated industries might have the scale or the position to be able to invest in AI innovation. Finally, any other efficiency-enhancing factor pushing both firms' sales and their innovative activity might be behind the observed relationship.

Table 4.5. Market concentration and patenting concentration

	Panel A				Panel B		
	(1)	(2)	(3)		(4)	(5)	(6)
Share by largest BG (patents, $PAT_{s,g,t}^{PL}$)	0.192*** (0.0704)	0.163** (0.0806)	0.167** (0.0831)	Share by largest BG (sales, $PAT_{s,g,t}^{ML}$)	0.426*** (0.152)	0.401** (0.155)	0.391** (0.162)
Observations	170	170	168		170	170	168
Adj. R ²	0.04	-0.03	-0.03		0.08	0.02	0.014
Controls	Tot. AI-pat	Tot. AI-pat	Tot. AI-pat		Tot. AI-pat	Tot. AI-pat	Tot. AI-pat
Fixed effects	No	Year	Year Geography		No	Year	Year Geography

Note: The dependent variable is CR4, market concentration. The regressions control for the yearly number of AI-related patent filings in each market and year and geography fixed effects. Standard errors are robust to heteroskedasticity. Significance level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. This sample includes 15 industries (mostly from Manufacture of basic pharmaceutical products; Manufacture of computer, electronic and optical products; Professional, scientific, and technical activities) over the years 2000-2019. Each industry is classified either as domestic, European, or global, according to the taxonomy by Calligaris et al. (2024_[66]). The countries included in the sample are Denmark, Finland, France, Germany, Greece, Hungary, Italy, Norway, Poland, Portugal, Slovenia, Spain, Sweden, and the United Kingdom for the domestic and European buckets, while in the global one also Japan, Korea, and the United States are included.

Source: OECD calculations based on PATSTAT Global and Moody's Orbis.

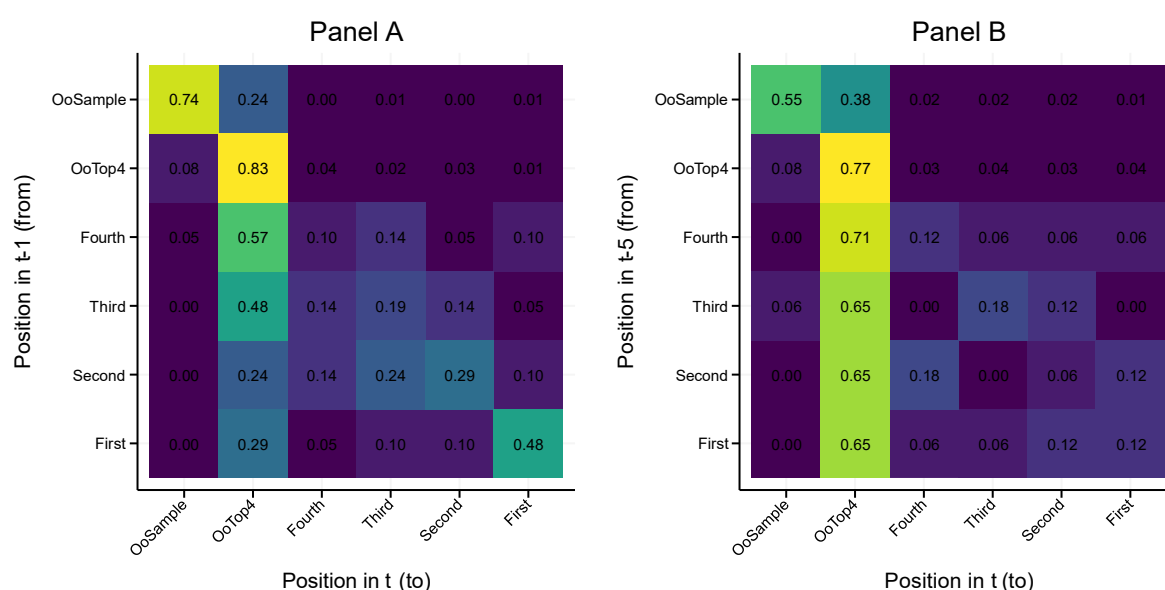
Dynamism among AI innovators

Another important dimension of competition in the age of AI is the dynamism of the competitive environment among AI innovators. Assessing the persistence in the ranking of top patentees can shed light on whether AI innovation is becoming increasingly entrenched among a stable group of dominant firms or remains open and competitive; this is important to understand possible future market developments.

In order to analyse these dimensions, firms are ranked by the number of AI-related patent filings every year and assigned to six categories: one for each of the top four positions; one for being out of the top four but still having an invention (OoTop4); and one that gathers all firms present in the sample not patenting in that year (OoSample). The analysis leverages on the flow rather than on the stock of filings. Using the stock would mechanically display a higher built-in persistence, while the focus is on the presence of the same firms at the frontier of AI innovation, which might be more erratic in the short run. Then, the probabilities of moving across categories are computed and the results are reported in Figure 4.6 using transition matrices. The matrix in Panel A reports the transition probabilities across the ranking from year $t-1$ to year t . From one year to the following, there seems to be contestability in the ranking, with a high probability of applying for relatively fewer (or zero) patents in year t than in $t-1$, for any position in the ranking (second column). The only ranking position showing a certain persistence is the first, with a probability of 0.48 or remaining in that position over two consecutive years. Panel B of Figure 4.6 takes a longer time horizon perspective and computes the transition probabilities across different positions between $t-5$ and t . Over a longer period of time the probability of remaining at the top of the ranking drops significantly while the probability of leaving the top four groups becomes relatively high (second column).

Overall, while in the very short run there is a certain persistence in the composition of the top four group, in the medium/long run there is a considerable reshuffling of the firms' relative positions in terms of AI-related invention flow. This suggests a relatively dynamic AI patenting environment.

Figure 4.6. Transition matrix of Top 4 firms in terms of yearly inventions.



Note: The two panels represent the transition matrices across positions of AI-related patentees. Panel A represents the transition probabilities from one year to the next, while Panel B takes a 5-years window perspective. Column labels refer to the position in the first period, while row labels refer to position in the second period. For a given period, OoSample refers to being out of the sample of patenting firms, while OoTop4 refers to being out of the top4 positions in patenting. Calculations are based on a sample gathering firms with AI-related patents that are also recorded in Moody's Orbis across 45 countries, with Canada, China, France, Germany, Israel, Japan, the Netherlands Korea, Switzerland, Chinese Taipei, the United Kingdom, and the United States, accounting for more than 90% of observations.

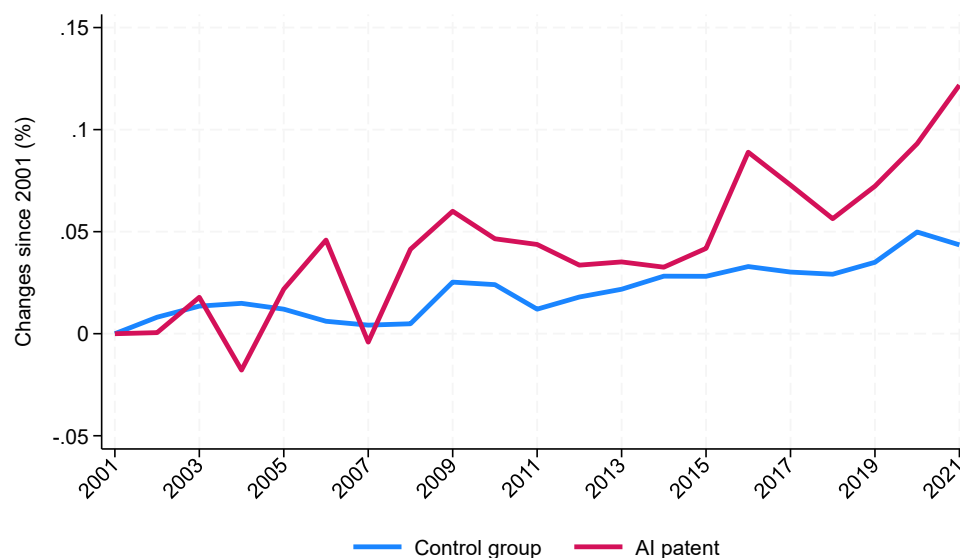
Source: OECD calculations based on PATSTAT Global and Moody's Orbis.

AI innovation and markups

At least conceptually, the relationship between patents and markups is well-grounded. By granting a temporary monopoly on an invention, a patent confers a certain degree of market power and, therefore, the possibility to raise prices above marginal costs. This mechanism might be particularly relevant for product innovations (notably the ICT sector inventing new algorithms/models) rather than for process innovation in non-ICT sectors, where AI innovations are complementary to firms' activities rather than being more closely related to the main one.

To meaningfully explore whether AI patenting endowed firms with market power, markups of firms with AI-related patents are compared with markups of firms holding other types of patents.⁵⁰ Figure 4.7 shows that, on average, markups of firms with AI-related patents had a slightly higher cumulated growth with respect to the control group. Interestingly, Figure A B.3 shows that this picture changes across ICT (broadly defined) and non-ICT sectors.⁵¹ In non-ICT sectors markups increased much less than on average, and AI-related patents seem not to provide market power. On the other hand, as shown in Panel B, ICT sectors display generally higher markup growth and a positive markup premium for firms associated to AI-related patents.⁵² A possible interpretation is that in ICT sectors AI inventions are actually related to the core business (either considering AI as an input, or as an output), and such proximity increases the relevance of AI-related patents in granting market power.

Figure 4.7. Average markup trend



Note: The lines in the figure represent the growth rate of average markups across two groups of firms. The blue line is based on a sample of patenting firms (non-AI-related), while the red one is based on firms associated with at least one AI-related invention, and only for the years after the first AI-related invention. Countries included in the sample are Austria, Belgium, Czech Republic, Denmark, Estonia, Finland, France, Germany, Hungary, Ireland, Italy, Latvia, Luxembourg, the Netherlands, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, and the United Kingdom.

Source: OECD calculations based on PATSTAT Global and Moody's Orbis.

The relationship between AI-related patents and markups is further explored in a sample of patenting firms through a firm-level regression framework, described by Equation 4:

$$\log(\mu_{i,t}) = \beta_0 + \beta_1 AI_{i,t} + \beta_2 ICT_i + \beta_3 AI_{i,t} * ICT_i + \beta X_{i,t} + \gamma_{s,c,t} + \varepsilon_{i,t}.$$

Equation 4

On the left-hand side, the logarithm of firms' markups ($\mu_{i,t}$) is regressed against an indicator that takes value equal to one after the first AI-related patent application associated with the firm ($AI_{i,t}$).⁵³ ICT_i indicates whether the firms belong to ICT sectors and $X_{i,t}$ is a vector of firm-level controls, which include capital intensity, size class, and age class. Finally, $\gamma_{s,c,t}$ is a series of different fixed effects which changes depending on the specification (see bottom of Table 4.6). Of particular interest are the estimates of parameter β_3 , which capture the additional markup premia associated with AI patents in ICT sectors.

Results in Table 4.6 show that on average firms with AI-related patents have slightly higher markups (column 1).⁵⁴ However, once ICT sector controls are included (columns 2-5), results do not hold: the coefficient becomes small or negative, implying that the raw positive association is largely driven by ICT sectors. In non-ICT sectors, AI patents are not associated with higher markups, and may even correlate with slightly lower markups (columns 3-4).

Further, the positive and significant interaction term in columns 3 and 4 suggests that, within ICT sectors, AI patents are associated with additional markup premia. This means that AI patents confer market power primarily in sectors where AI is strategically important. This relationship is robust across specifications using different fixed effects, although it vanishes when using firm fixed effects. This is likely due to two related issues. First, the $AI_{i,t}$ variable has a low temporal variation available for the model to explore once firms adopt AI, as shown in Figure A B.4. Second, the fixed effects absorb the unobserved quality of the

most innovative firms, which are likely to be the earliest adopters, making it difficult to separate the effect of the technology from the firm's inherent characteristics.⁵⁵ The figure shows how a lot of AI patenting occurs late in the sample, and this might prevent market power associated with AI-related patents to materialise.

Overall, the results suggest that AI patents are positively related with markups mainly in sectors where AI is closer to the firms' core business, rather than an additional input in the production process. Outside ICT industries, the presence of AI patent applications is not associated with higher markups. This suggests that patent filings that are most likely linked to activities that are complementary with the firms' main ones are not associated with market power.

Table 4.6. AI patenting and firms' markups

	(1)	(2)	(3)	(4)	(5)
AI	0.0376*** (0.0076)	0.0018 (0.0073)	-0.0370*** (0.0080)	-0.0565*** (0.0082)	-0.0144 (0.0103)
ICT		0.1848*** (0.0019)	0.1754*** (0.0019)		
AI x ICT			0.0795*** (0.0174)	0.0784*** (0.0173)	0.0055 (0.0239)
Observations	606421	606421	606408	593141	601747
Adj. R ²	0.084	0.11	0.13	0.16	0.73
Controls	Yes	Yes	Yes	Yes	Yes
Fixed effects	No	No	Cou-year	Cou - Year, Ind-Year	id year

Note: The dependent variable is the logarithm of firms' markups. AI is a dummy equal to one from the first year a firm becomes associated with an AI-related invention. ICT is a dummy indicating whether an industry belongs to the ICT sector, broadly defined, and including Manufacture of computer, electronic and optical products (26), Publishing, audiovisual and broadcasting activities (58 to 60), Telecommunications (61), IT and other information services and (62 to 63). Further controls include age class, size class, dummy or the firm being at the TFP frontier, and capital intensity. Standard errors are robust to heteroskedasticity. * p<0.10, ** p<0.05, *** p<0.01. Countries included in the sample are Austria, Belgium, Czech Republic, Denmark, Estonia, Finland, France, Germany, Hungary, Ireland, Italy, Latvia, Luxembourg, the Netherlands, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, and the United Kingdom.

Source: OECD calculations based on PATSTAT Global and Moody's Orbis.

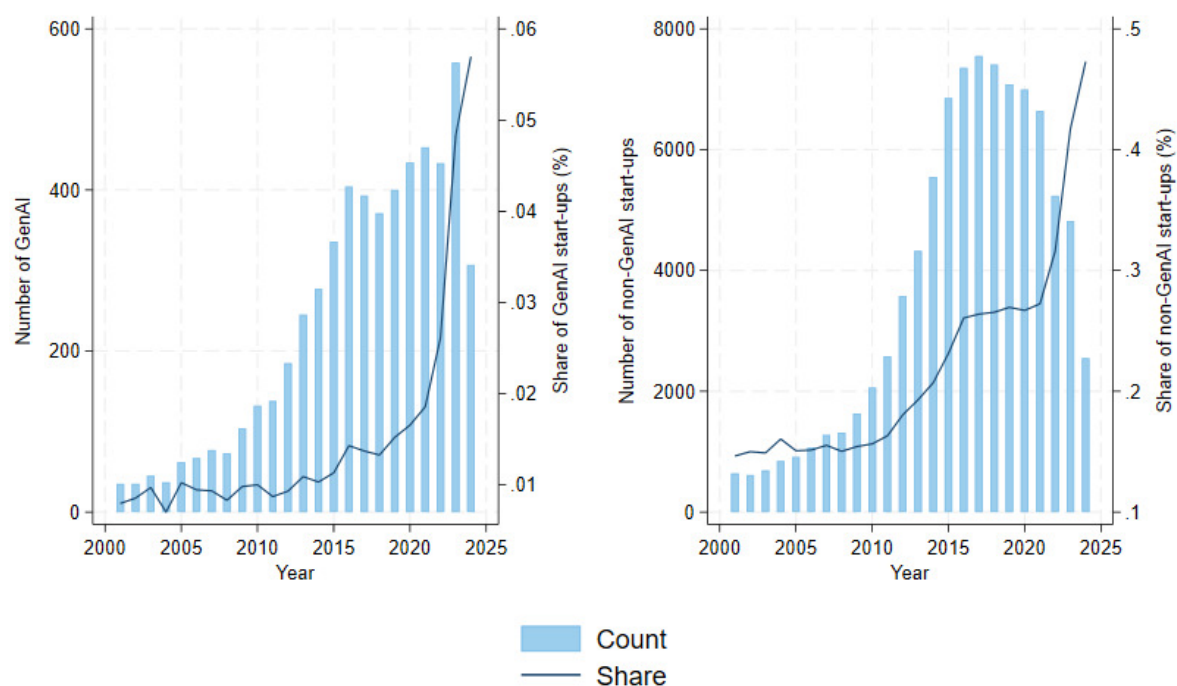
Evidence on the AI start-ups ecosystem

Start-ups provide a unique lens through which to study the implications for competition of emerging technologies such as artificial intelligence. Young innovative firms are often responsible for the most radical technological breakthroughs, and their entry, growth, and exit dynamics are core components to foster competition (Fons-Rosen, Roldan-Blanco and Schmitz, 2021^[65]; Gans, Hsu and Stern, 2002^[75]). They contribute to market dynamism by experimenting with new business models, challenging incumbents technologically, and – when successful – reshaping industry dynamics. Understanding how start-ups engage with AI technologies therefore offers valuable insights into the trajectory of innovation, the potential for creative destruction, and the future structure of markets.

Figure 4.8 show a remarkable increase in the number and share of both non-GenAI and GenAI start-ups, in panel A and B respectively. The left axis in both panels reports the absolute counts over the past decade, while the right axis shows the *share* of AI-related start-ups. The recent decline in absolute counts (post-2021) is not indicative of a slowdown in AI entrepreneurship. Rather, it reflects well-known data lags in the registration and classification of very young start-ups, which affect the most recent founding cohorts in the OECD Start-up Database. In contrast, the *share* of non-GenAI and GenAI start-ups – much less sensitive to these mechanical lags – continues to rise steeply. This suggests that AI-related entrepreneurship is

expanding rapidly relative to other technological domains, with GenAI in particular exhibiting exponential growth since 2020.

Figure 4.8. Number and share of and GenAI and non-GenAI start-ups by foundation year

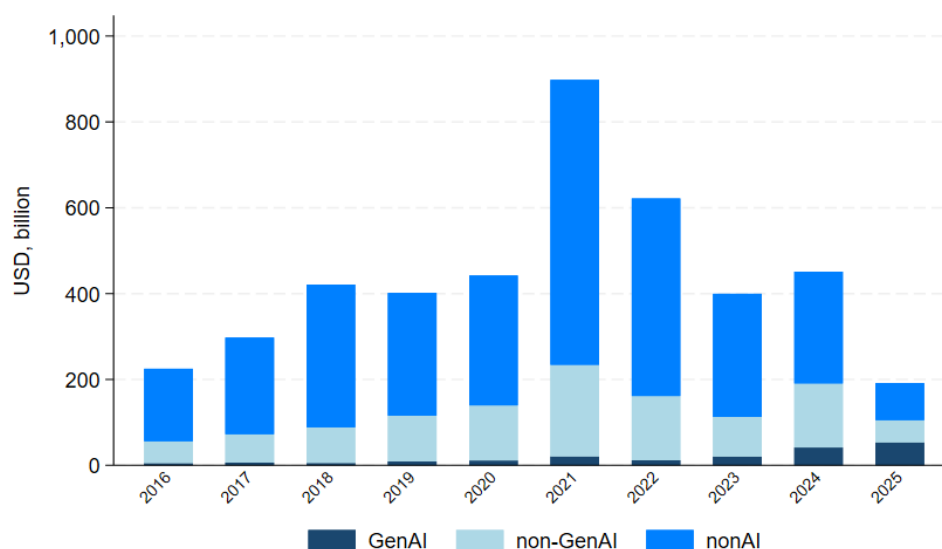


Note: The bars report the number of GenAI (left figure) and non-GenAI (right figure) start-ups by foundation year. The line reports the share of GenAI (left figure) and non-GenAI (right figure) start-ups by foundation year relative to the total number of start-ups founded in each year. The sample of start-ups is restricted to VC-backed or patenting start-ups.

Source: OECD calculations based on OECD Start-ups Database.

Figure 4.9 further shows that VC investment has increasingly shifted toward AI-related start-ups. Across the entire period, AI start-ups attract a growing share of total VC funding, with GenAI emerging as a distinct and rapidly expanding category after 2020. This pattern persists despite the overall cooling of global VC markets after the 2021 peak. VC financing matters not only as a measure of investor expectations, but also because scaling up AI technologies requires significant early capital investment – for model development, compute resources, and specialised talent. The fact that large and growing VC flows are channelled toward AI and GenAI start-ups indicates that investors perceive these technologies as having high commercial potential and substantial transformative impact across sectors. This inflow of capital can accelerate the expansion of young AI firms, potentially enabling them to challenge incumbents or create entirely new markets.⁵⁶

Figure 4.9. VC funding by year across types of start-ups



Note: The figure reports the amount of VC equity financing expressed in USD billions by start-ups in the OECD start-up database.
Source: OECD calculations based on OECD Start-ups Database.

Going beyond descriptive evidence, a simple OLS regression is performed to assess whether GenAI and non-GenAI start-ups are indeed more likely to raise more VC funding compared to other start-ups. From a competition standpoint, higher VC funding for GenAI and non-GenAI start-ups may have important implications for market dynamics, by affecting entry, scaling, and the competitive pressure these firms can exert on incumbent players. The baseline estimating equation is the following:

$$\log(VC\ funding_{f,c,t}) = \beta_0 + \beta_1 GenAI_{f,c,t} + \beta_2 StatAI_{f,c,t} + \beta X_{f,c,t} + \varepsilon_{f,c,t},$$

Equation 5

Where $VC\ funding_f$ reports the (log of) total VC raised by a start-up, while $GenAI_f$ and $StatAI_f$ are firm-level dummy variables indicating whether an AI start-up is classified as GenAI or non-GenAI, respectively. The objective of this specification is to understand whether GenAI or non-GenAI start-ups receive more VC funding than nonAI start-ups, the reference category.

Table 4.7. Determinants of Total VC funding raised by start-ups

	(1)	(2)	(3)	(4)	(5)	(6)
Gen AI	2.219*** (0.0959)	1.916*** (0.0951)	2.217*** (0.0959)	1.473*** (0.106)	1.522*** (0.106)	1.314*** (0.134)
Non-Gen AI	0.818*** (0.0280)	0.683*** (0.0279)	0.818*** (0.0280)	0.465*** (0.0325)	0.482*** (0.0326)	0.455*** (0.0535)
Log Tot patents			0.695*** (0.0463)			0.677*** (0.0610)
Has serial founder				0.925*** (0.0349)		1.881*** (0.0831)
Has PhD founder					0.557*** (0.0472)	0.535*** (0.0753)
Obs.	382 108	381 509	382 108	249 441	249 441	248 905
Adj. R ²	0.00331	0.0661	0.00406	0.00451	0.00233	0.0626
Cou-Year FE	No	Yes	No	No	No	Yes

Note: the table reports the results of Equation 5. The dependent variable is the log of the total amount of VC raised by a start-ups over its lifecycle. The main explanatory variables are two dummy variables, Gen AI and Non-Gen AI, which are 1 if a start-up is considered as Gen AI or Non-Gen AI, respectively, and 0 otherwise. Other control variables included in the different columns are the log of 1 + the total number of patents filed before the first VC financing, a dummy variable indicating whether the start-up is founded by at least a serial founder (i.e. a founder finding more than one start-up in the OECD Start-up database) and a dummy variable indicating whether the start-up is founded by at least a founder with a PhD degree. Columns 2 and 7 include country-year of foundation fixed effects. Standard errors are robust to heteroskedasticity. * p<0.10, ** p<0.05, *** p<0.01.

Source: OECD calculations based on OECD Start-ups Database.

Across different model specifications, additional control variables are progressively included to examine whether the relationship between AI type (GenAI, non-GenAI, or nonAI) and VC funding depends on other observable characteristics. Column 2 of Table 4.7 includes country *c* and year-of-foundation *t* fixed effects to account for unobserved heterogeneity across countries and founding cohorts (e.g. the VC boom in the US in 2021). Column 3 adds a control for the log of the number of patents filed by the start-up at the time of the first VC round.⁵⁷ Column 4 controls for whether the start-up was founded by at least one serial entrepreneur, i.e. founded by at least a founder that had founded more than one start-up, and column 5 includes a dummy for whether at least one founder holds a PhD. Finally, column 6 includes all these control variables jointly.

As reported in the table, AI-related start-ups receive substantially more VC funding than nonAI start-ups. The relationship is robust across all specifications, even after controlling for several firm-level or founder characteristics typically associated with VC financing. This suggests that VC markets are currently highly enthusiastic about AI developments and are allocating disproportionately large amounts of capital to AI start-ups. The effect is particularly pronounced for GenAI start-ups. For example, the coefficient in column 6 indicates that GenAI start-ups receive, on average, approximately 130% more VC funding than otherwise similar nonAI start-ups.

A similar pattern emerges when considering the broader set of funding received by start-ups, beyond VC instruments.⁵⁸ Relevantly, the OECD Start-ups database distinguishes funding by stage of development (seed, early, growth, and late stage), allowing the analysis to be conducted across different points in the financing lifecycle. The results remain robust across all stages, although the relationship is particularly strong for seed and early-stage funding. This is not surprising. A large share of AI-related start-ups has been founded only in recent years, meaning that these firms are disproportionately represented among early-stage cohorts. Moreover, early-stage investors may be especially responsive to emerging technological trends, making them more willing to allocate capital to AI-related projects anticipating a future rapid market expansion.

While the evidence reported in this subsection depicts a dynamic AI-related entrepreneurial ecosystem, funding on its own is not a sufficient condition for promising start-ups to grow and scale – ultimately becoming tomorrow’s potential competitors to today’s incumbents and realising the process of creative destruction. For example, another important dimension to foster start-up growth, is represented by a well-functioning exit market. In the entrepreneurial literature, exit opportunities have long been considered a central mechanism because they determine investors’ ability to realise returns and reinvest capital into new ventures, thereby sustaining the broader innovation cycle (Gompers and Lerner, 1999^[76]; Lerner, 1994^[77]; Nanda and Rhodes-Kropf, 2013^[78]). Traditionally, IPOs were the dominant exit route (Rhodes-Kropf and Robinson, 2004^[79]). In recent years, however, the balance has shifted markedly toward acquisitions (Berger et al., 2025^[30]). This shift can foster entry and support VC activity by providing credible and timely exit options (Fons-Rosen, Roldan-Blanco and Schmitz, 2021^[65]). Yet, the increasing reliance on acquisitions as the primary exit channel also raises concerns from a competition perspective, particularly when dominant incumbents acquire emerging rivals, potentially dampening future innovation and limiting the emergence of competitive threats (Berger et al., 2025^[30]; Cunningham, Ederer and Ma, 2021^[80]).

To provide a better understanding of how the AI entrepreneurial ecosystem may affect the relationship between AI and competition, the analysis also focuses on the likelihood of acquisition for AI-related start-ups. To do so, the following linear probability model is estimated:

$$1\{\text{Acquisition}\}_{f,c,t} = \beta_0 + \beta_1 \text{GenAI}_{f,c,t} + \beta_2 \text{StatAI}_{f,c,t} + \beta X_{f,c,t} + \varepsilon_{f,c,t},$$

Equation 6

where the dependent variable is an indicator equal to one if the start-up experiences an acquisition exit. To ensure that start-ups have had a sufficient exposure window for acquisition, the analysis restricts the sample to firms founded before 2024. This restriction improves comparability across AI and non-AI start-ups and reduces bias arising from differences in firm age or cohort maturity. The specification progressively introduces controls that capture both observable characteristics at the time of exit (e.g. accumulated funding, patent stock), founder attributes and country c and founding year t fixed effects, allowing to assess how the association between AI types and acquisition dynamics evolves as richer information is included.

Table 4.8 reports the results. Across specifications, GenAI start-ups consistently display a higher probability of being acquired. The baseline coefficient (column 1) indicates that GenAI firms are about 1.6 percentage points more likely to be acquired, and this relationship remains statistically significant even after introducing country-year fixed effects (column 2) or controlling for start-up characteristics at exit (columns 3-7). Regarding economic magnitude, the unconditional probability of acquisition in the sample is 7.58%.⁵⁹ This implies that GenAI start-ups exhibit an acquisition premium of roughly 21% relative to the average start-up. This pattern suggests that GenAI businesses are particularly attractive acquisition targets – possibly reflecting their rapid commercial scaling potential, strategic relevance for incumbents, or complementarities with established firms’ AI capabilities.

Table 4.8. Determinants of start-up acquisition

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
GenAI	0.0161*** (0.00402)	0.0342*** (0.00681)	0.0109*** (0.00400)	0.0148*** (0.00400)	0.0158*** (0.00503)	0.0127** (0.00503)	0.0332*** (0.00710)
Non-GenAI	0.000369 (0.00103)	0.0146*** (0.00263)	-0.00146 (0.00103)	-0.000633 (0.00103)	-0.000907 (0.00138)	-0.00234* (0.00138)	0.0157*** (0.00254)
Tot funding at exit			0.00298*** (0.0000639)				0.00285*** (0.000243)
Tot patent at exit				0.0241*** (0.000689)			-0.0123*** (0.00161)
PhD founder					0.0159*** (0.00199)		-0.0000108 (0.00189)
Serial founder						0.0424*** (0.00128)	0.0503*** (0.00388)
Obs.	376 305	375 758	376 305	376 305	245 977	245 977	245 481
R ²	0.0000505	0.0573	0.00636	0.00508	0.000340	0.00345	0.0865
Cou-year FE	No	Yes	No	No	No	No	Yes

Note: the table reports the results of Equation 6. The dependent variable is a dummy variable equal to 1 for acquired start-ups, 0 otherwise. The main explanatory variables are two dummy variables, Gen AI and Non-Gen AI, which are 1 if a start-up is considered as Gen AI or Non-Gen AI, respectively, and 0 otherwise. Other control variables included in the different columns are the log of 1 + the total number of patents filed before the acquisition, the total amount of VC raised before the acquisition, a dummy variable indicating whether the start-up is founded by at least a serial founder (i.e. a founder finding more than one start-up in the OECD Start-up database) and a dummy variable indicating whether the start-up is founded by at least a founder with a PhD degree. Columns 2 and 7 include country-year of foundation fixed effects. Standard errors are robust to heteroskedasticity. * p<0.10, ** p<0.05, *** p<0.01.

Source: OECD calculations based on OECD Start-ups Database.

For start-ups developing AI technologies more broadly (non-GenAI), the relationship is weaker and more sensitive to controls. The coefficient becomes sizable and positive only in the specifications including country–foundation year fixed effects (column 2 and 7). Given that there has been considerable entry of AI start-ups, this control seems particularly important given that younger start-ups are mechanically less likely to be acquired simply because they have had less time to mature and be targeted by incumbents. By including these fixed effects, the analysis compares start-ups within the same country and founding cohort, thereby isolating differences in acquisition likelihood that are not driven by age or broad macro trends. Under this more appropriate comparison, non-GenAI start-ups display a positive and significant association with acquisition probability, although smaller in magnitude than the effect estimated for GenAI firms.

Overall, these results show that GenAI start-ups are more likely to be acquired, even conditional on funding, patents, founder experience, and macro factors.⁶⁰ This underscores the strong strategic importance attributed to GenAI technologies by incumbent firms, and aligns with the broader shift toward acquisition-led exits observed in the entrepreneurial ecosystem. Although acquisition exits can foster start-up entry and growth (Fons-Rosen, Roldan-Blanco and Schmitz, 2021^[65]), the tendency for GenAI start-ups to be acquired more frequently may warrant attention regarding potential effects on competition and future innovation developments. In particular, the pattern may raise concerns about whether these acquisitions spur technology diffusion and integration – where the acquired firm's innovations reach broader markets through the acquirer's scale – or rather strategic consolidation, where incumbents acquire emerging rivals primarily to neutralise competitive threats. The latter dynamic is consistent with so-called “killer acquisitions”, where the acquirer discontinues the target's innovation to protect existing market positions (Berger et al., 2025^[30]; Cunningham, Ederer and Ma, 2021^[80]).

Linking the initial empirical evidence to the conceptual framework

Building upon the evidence presented above, this section aims at linking the initial empirical results to the conceptual framework discussed in Section 2, helping draw a first picture of the relationship between AI and competition in its different facets. As previously mentioned, given data availability, the analysis did not focus specifically on competition dynamics among foundation models – which are object of complementary analysis (André et al., 2025^[27]) – but rather focused on more downstream analysis. Also, the focus remained more closely focused on AI itself rather than on the broader AI value chain (see OECD (2025^[23]) for further discussion).

Despite these caveats, leveraging information from different sources of microdata across countries – notably ICT surveys on AI use, LEED data on occupational exposure to generative AI, patent data identifying AI-related innovations, as well as AI start-up data – has allowed providing a uniquely broad overview of different facets of the links between AI and competition in the age of AI.

Analysis of firm-level ICT surveys has provided evidence that appears mainly relevant for quadrant of non-generative AI users. In fact, the proxy employed in those estimates and the timing of the available data do not allow to comprehensively capture the recent boom in generative AI use. While the presented evidence should be interpreted with caution, due to the short time period considered and the lack of a causal identification strategy, it suggests that – on average and in the period under analysis – AI use by firms does not seem related to substantial increases in the examined proxies of market power in France and Portugal. This is particularly true for markups trends, which do not appear significantly different between AI users and other firms, while for market shares dynamics some specifications point to a possible role of AI strengthening the edge of most competitive firms. This reinforces the relevance of continuing monitoring such dynamics in the future.

Focusing more closely on (potential) Generative AI users (the top left quadrant), analysis of LEED provided further relevant insights. On the one hand, generative AI may represent a relevant opportunity for smaller and younger firms, which may be well positioned to adopt AI and challenge market leaders. On the other hand, the benefits of generative AI may be unlocked more easily by firms that have higher productivity and capabilities, notably including workforce skills. This may highlight risks of increasing divides between the most productive firms and the rest of the business population. For laggard firms, benefiting from recent developments in AI may require significant changes, in particular in terms of human capital. If AI quickly spreads as a general-purpose technology, a significant share of the economy may not be able to fully extract its benefits and could potentially be driven to lower performance or market exit.

Evidence from AI patents appear relevant to assess the dynamics of (a subset of) AI developers, particularly in relation to the non-generative AI developers quadrant (although some of the patents may also relate to the development of Generative AI systems). The results show mixed patterns in the concentration of AI-related patenting. Although concentration rose after 2010, it declined over a longer horizon, mainly reflecting the growing role of medium-sized innovators rather than reduced activity by the largest players. At the same time, concentration in AI patenting is positively correlated with market concentration, which may reflect greater advantages gained by market leaders, a better capacity of firms in concentrated industries to invest in AI, or broader efficiency gains of market leaders pushing both AI adoption and rising concentration. Patenting activity also appears dynamic, with some short run persistence but significant medium- to long run turnover among top filers. Finally, after the filing of their first AI-related patent, firms exhibit higher markup growth, but only in ICT sectors. This may suggest that AI may contribute to market power mainly where it represents an output rather than an input for firms.

A different facet of the links between AI and competition is offered by information on non-generative and generative AI start-ups. While it is not straightforward to link such evidence to either the development or use dimensions of the conceptual framework, one may think that start-ups may be closer to the development side, as they often bring to the market new technologies and – given their identification based

on tags, business descriptions and patents – they likely have AI at the core of their business. The evidence shows a relevant increase in the number and shares of both non-GenAI and GenAI start-ups, with VC investments increasingly shifting toward AI-related start-ups. Results further show that GenAI start-ups are disproportionately more likely to be acquired, even conditional on funding, patents, founder experience, and macro factors. This underscores the strategic importance attributed to GenAI technologies by incumbent firms and aligns with the broader shift toward acquisition-led exits observed in the entrepreneurial ecosystem. Although the potential of being acquired can foster start-up entry and growth, this tendency may warrant attention regarding potential impacts on competition and future innovation developments.

Three caveats are important when interpreting these dynamic results. First, the findings relate to a relatively short time horizon and the gains from foundational investments like AI often follow a J-curve effect, with costs preceding benefits. It is therefore possible that significant, longer-term effects on the proxies of competitions have not yet had time to fully materialise. Second, the mixed results could hide important heterogeneity. For example, AI adoption may be positively related to market power gains for some firms (e.g. specific types of adopters or those in certain parts of the market share distribution) but negatively related for others. Exploring this potential heterogeneity is a promising avenue for future research. Third, across the different exercises, the correlational nature of the analysis means that reverse causality remains a possibility – for instance, firms with pre-existing market power may be better positioned to invest in and adopt AI, possibly contributing to the observed associations.

5 Conclusions

This report presents initial evidence on the relationship between AI and competition. It introduces a novel conceptual framework that classifies different types of AI and actors in the AI ecosystem. The framework distinguishes between non-generative and generative AI tools, and between two groups of AI-involved firms: AI users and AI developers. By combining these dimensions, the framework identifies four different categories of relationships and helps elicit the possible channels through which AI can affect the evolution of competition.

Building on this framework, the report leverages a broad and diverse set of data sources to provide early empirical evidence on selected facets of the AI-competition nexus. Overall, this paper provides a mixed, rather than an alarming, picture. Over the period under analysis, non-GenAI use by firms does not seem consistently related to significant increases in markups and market shares, and the AI development landscape itself displays some degree of dynamism. At the same time, some results indicate that non-GenAI may reinforce the competitive edge of firms already at the top of the market share distribution and may provide some market power, pointing to the possibility of heterogeneous effects and, possibly, concerns for the long run. While the analysis is limited by data availability, which does not always capture the relatively recent AI diffusion and developments, it establishes a first empirical benchmark that can be used to monitor the evolution of such a relationship.

These findings have several policy implications. First, these initial mixed results on short-term trends warrant further monitoring from policymakers. The absence of systematic markup and market share growth suggests that *ex-ante* regulation to curb possible harms might be, at least at this stage, premature. However, competition authorities should closely monitor possible market consolidation moves by the largest players, in particular in the acquisition of innovative and disruptive AI developers, to ensure that the current dynamism and competition in AI development are maintained.

Second, the evidence that the GenAI development landscape is relatively dynamic points towards policies that continuously support innovation and lower barriers to entry. In particular, access to key inputs such as computational power and skilled labour should be a priority for start-ups both looking to develop and to adopt AI. The heterogeneity in the share of the workforce with tertiary education between firms with different levels of GenAI exposure also reinforces the need for continued investment in skills. If the effective use of AI requires a highly skilled workforce, then barriers to accessing AI-relevant skills may effectively become barriers to entry in AI-intensive markets. In this sense, labour market and skills policies could have direct implications for competition.

Third and finally, the rapid pace of AI diffusion means that available data do not yet fully capture recent developments, particularly concerning GenAI. Continued efforts by national statistical organisations to collect more comprehensive and granular data on firm-level AI adoption and development would be valuable for monitoring the relationship between AI and competition. In particular, data that distinguish between different types of AI technologies and AI actors, capture the type and intensity of use, and allow tracking adoption events and dynamics over time would strengthen the ability to assess competitive dynamics as AI diffusion progresses.

The analysis presented in this report opens several avenues for future research. On the AI use side, a natural extension would involve moving beyond the binary adoption indicator employed in the ICT survey

analysis to measures of AI intensity, or by linking outcomes to each AI technology type that make up the broad binary adoption indicator, as well as the distinction between AI developed in-house and purchased from external providers. Further work could also exploit the panel structure of ICT surveys to track changes in adoption status over time. As for the analysis of occupational exposure to GenAI, a relevant next step may be to extend it beyond descriptive evidence, for example by directly relating exposure measures to competition outcomes in a regression framework and interacting exposure with adoption indicators. The analysis on start-ups acquisition could also be further extended by distinguishing among types of acquirers, which would help assess the extent to which acquisitions reflect strategic consolidation as opposed to technology diffusion. More broadly, next steps extending the country coverage of the firm-level analysis and further distinguishing between generative and non-generative AI across empirical exercises could broaden the generalisability and granularity of the findings.

Endnotes

¹ According to Jovanovic and Rousseau (2005^[92]), General Purpose Technologies (GPT) have three main characteristics: pervasiveness, by spreading across most sectors; improvement, as they should get better over time; innovation spawning, by facilitating the invention of new products or processes.

² Broadly speaking, industry concentration captures the share of output accounted for by the largest firms within an industry, while markups measure the wedge between a firm's output price and its marginal cost, providing the most direct measure of market power.

³ The increased importance of intangibles might have also influenced markup dynamics. Models *à la* De Ridder (2024^[16]) suggest that large firms incur the fixed cost of investing in intangible assets, which in turn leads to a reduction in marginal costs, thereby increasing markups.

⁴ See Calvino and Criscuolo (2019^[96]) and Calvino, Criscuolo and Verlhac (2020^[12]) for related discussion about business dynamics and digitalisation.

⁵ In this sense, it focuses more on the downstream layers of the AI production chain such as the model development and the use by companies, neglecting upper layers such as the hardware or the cloud components (Hagiu and Wright (2025^[20]; André et al., 2025^[27]). While important, the study of these segments of the AI stack goes beyond the scope of the current work.

⁶ The framework can be further used to investigate how competition among market leaders can shape AI development and its adoption by firms.

⁷ A more precise terminology would distinguish between statistical non-generative approaches and symbolic approaches within non-generative AI. For simplicity, the term “non-generative AI” is used throughout and refers to statistical non-generative AI applications.

⁸ Although, most of the time, the two terms are used interchangeably, it is helpful to distinguish between foundation models and Large Language Models (LLM). The former are large AI models that can be adapted to different applications. LLMs are a particular (and the most common) type of foundation models focused on language recognition and generation.

⁹ A few typical examples include the automation of some repetitive tasks within occupations and the augmentation of others; applications related to customer services; strategic, logistic and sales analysis; product innovation; legal and risk analysis; coding and software development.

¹⁰ See Calvino, Reijerink and Samek (2025^[32]) for further discussion in a recent review of the experimental literature on the effects of generative AI on productivity, innovation and entrepreneurship.

¹¹ Apart from standard tools that do not need any adaptation for the purposes of companies employing them, a minimum amount of customisation of further development might be required by some applications. For this reason, a considerable part of users could also be considered as developers.

¹² Through upstream integration, AI developers can ensure a steady flow of essential data and technology, further strengthening their market position. Through downstream integration, and thus by integrating directly with end users, AI developers can create tailored solutions that meet specific market needs, thereby enhancing their product offerings and solidifying their market presence. These dynamics raise concerns about possible lock-in effects for users of integrated systems: firms using bundled AI solutions may become reliant on specific providers, creating long-term dependencies that hinder innovation and reduce competition (OECD, 2024^[93]; OECD, 2025^[23]).

¹³ There are at least three categories: models derived from larger foundation ones as MobileBERT, derived from Google's BERT; light, independent models developed by companies offering larger models, as MiniLM by Microsoft Research; models by small independent firms with, usually, more targeted purposes: Aleph Alpha is an independent company focusing on specific business problems and maintaining adherence towards certain data protection standards, while e.g. Cohere provides light models specifically suited for SMEs.

¹⁴ A key distinction to consider is between open-source and proprietary AI models. The dominance of proprietary models can create lasting barriers to adoption by firms, granting permanent advantages to early adopters and reinforcing their market power.

¹⁵ Two possible examples of such relatively small companies are DataRoot Labs and Neurala.

¹⁶ A manufacturing company might decide to develop its own AI-based anomaly detection system to automatise its quality-control procedure.

¹⁷ The data at hand do not allow to zoom in on competition dynamics among foundation models – which are object of complementary analysis (André et al., 2025^[27]). Also, the focus is on AI itself rather than on the broader AI value chain (see OECD (2025^[23]) for further discussion).

¹⁸ This analysis complements recent OECD work on human capital as driver of AI adoption (e.g. Calvino, Costa and Haerle (2026^[98])). While that work identified the characteristics of firms that use AI, and the role of occupational composition and educational attainments, the current analysis focuses on the potential impact of GenAI by mapping the overlap between AI capabilities and workers' tasks.

¹⁹ The study does not explicitly distinguish between patents in GenAI and non-GenAI. However, given the time span covered by the available data at the time of the analysis, this omission is less concerning, as GenAI substantially emerged only later.

²⁰ A point worth considering is that the framework, while static – it does not explicitly include a time dimension – can nevertheless be useful for understanding, for each quadrant, patterns in the short and long term. A second important caveat is that the competition conditions in one quadrant might affect those in other quadrants as well. While investigating these linkages goes beyond the scope of this work, this

aspect strongly highlights the importance of monitoring the competition conditions in relation to each technology-user type, as they might affect economic incentives in other quadrants of the framework.

²¹ An opportunity provided by GenAI is also the ability to reach new markets, for example by easily customising marketing campaign in other languages.

²² In an AI model, broadly speaking, training data are needed to create and adjust the prediction algorithms. Input data, instead, are used to obtain accurate predictions from the algorithms. The ownership of training data, despite their non-rivalrous nature, becomes a potential barrier to entry, as much as other AI-related resources such as compute. Input data, by enhancing firms' efficiency through AI algorithms, can be considered a driver of competitiveness.

²³ The distinction among model-providers is not always clear-cut though. Some non-frontier LLMs are often derived from larger models and have narrower applicability, but they are more portable and require fewer computational resources. On the other hand, even small independent companies may also provide such models for more targeted audiences.

²⁴ The different degrees of competition are described theoretically (from the most to the least competitive environment) as Bertrand with homogenous products, Cournot with homogenous product, Bertrand with product differentiation, and Cournot with product differentiation.

²⁵ Bertrand competition refers to a situation characterised by firms competing on price, with the most efficient firm eventually capturing the whole market.

²⁶ For France, see: https://ec.europa.eu/eurostat/cache/metadata/EN/isoc_e_simsie_fr.htm. For Portugal, see: https://ec.europa.eu/eurostat/cache/metadata/EN/isoc_e_simsie_pt.htm.

²⁷ While these surveys capture both the use of internally developed AI systems and the use of externally purchased ones, they do not explicitly distinguish between firms that develop AI for internal use versus for commercial sale.

²⁸ See the following links for more information: <https://www.casd.eu/source/statistique-structurelle-annuelle-dentreprises-issue-du-dispositif-suse/> and <https://www.casd.eu/en/source/annual-structural-statistics-of-companies-from-the-esane-scheme/>

²⁹ For an AI-related invention to be patentable, it must have a technical character, meaning that it uses technical means to address a technical problem. Software that merely implements or applies a machine-learning model is not patentable in itself, unless it makes a novel and non-obvious technical contribution to the solution of a technical problem (https://www.epo.org/en/legal/guidelines-epc/2025/g_ii_3_3_1.html, accessed on January 16th, 2026).

³⁰ A detailed discussion of the methodology developed to reconstruct the business groups, on the process of cleaning financial variables, as well as on the procedure to compute gross output of business group starting from affiliates specific unconsolidated sales, is available in Calligaris et al. (2024^[10]) and Bajgar et al. (2023^[11]).

³¹ Competition authorities define the relevant market along two dimensions. The product market dimension, comprising the products regarded as interchangeable or substitutable by the consumer by reason of the products' characteristics, their prices and their intended use. The geographic dimension, which is defined

by the area in which the firms concerned are involved in the supply of products or services and in which the conditions of competition are sufficiently similar (European Commission, 1997^[95]).

³² For further methodological details, see the Community survey on ICT usage and e-commerce in enterprises. Retrieved from https://ec.europa.eu/eurostat/cache/metadata/en/isoc_e_esms.htm#relatedmdDisseminated

³³ Earlier generative AI applications, including e.g. natural language generation, are however part of the AI adoption measure used for these estimates.

³⁴ The data sources used in both countries are highly representative, allowing precise computations of aggregate industry output and, in turn, market shares at firm level.

³⁵ For the analysis of markups, the dynamic variable is defined as the log-difference: $Y_{i,t} = \log(\mu_{i,t+2}) - \log(\mu_{i,t-3})$. For market shares, $Y_{i,t}$ reflects the direction of change in the firm's sector-specific ranking between $t - 3$ and $t + 2$. It takes the value 1 if $Decile_{i,t+2} > Decile_{i,t-3}$, -1 if $Decile_{i,t+2} < Decile_{i,t-3}$, and 0 otherwise.

³⁶ The AI indicator is a binary variable taking the value of 1 if a firm reports using at least one of the seven AI technologies surveyed, and 0 otherwise.

³⁷ The use of a binary AI adoption indicator may introduce attenuation bias. By grouping together firms that use a single AI application with those that deploy multiple technologies intensively, the binary measure dilutes the treatment group with firms for which AI use may be too limited to meaningfully affect competitive outcomes. This blurs the distinction between treated and untreated firms, pushing estimated coefficients toward zero. The null findings reported here could therefore partly reflect this measurement limitation.

³⁸ This is also true when focusing on actual market shares rather than deciles, as evident from unreported additional analysis.

³⁹ The heterogeneity in the links between AI use and market shares' growth is further explored using a quantile regression framework. While the baseline OLS estimates capture the average impact, quantile regressions allow us to examine how AI use relates to market share dynamics at different points of the distribution. Unreported preliminary evidence for Portugal suggests that, while AI use has no significant impact for the median firm, it is positively and significantly associated with market share growth for firms at the top of the distribution (i.e. those already experiencing the highest growth). This reinforces the relevance of continuing monitoring such dynamics in the future.

⁴⁰ Further unreported analysis suggests a limited degree of heterogeneity in the links between AI and markup growth across the markup growth distribution.

⁴¹ France and Portugal differ in their economic structures, firm size distributions, and levels of digital maturity. These structural differences may influence the competitive dynamics observed. As such, the findings from these two countries may not directly generalise to economies with substantially different competitive landscapes, industrial compositions, or levels of AI diffusion.

⁴² For example, Chatterji et al. (2025^[94]) documents the unprecedented speed of ChatGPT's adoption, which attracted over one million users within just five days of its launch in November 2022. This fast pace of adoption was sustained, reaching over 700 million weekly active users by July 2025. The volume of

interactions also continued growing, with users sending over 2.5 billion messages daily, and the total message volume growing by more than five times in the year between July 2024 and July 2025 alone.

⁴³ Felten, Raj and Seamans (2023^[58]) provide scores for both language modelling and image generation for each 5-digit SOC occupation. For the purpose of this exercise, these scores are averaged, and the resulting measure is then converted to 4-digit ISCO-08 using a crosswalk provided by the Bureau of Labor Statistics.

⁴⁴ This relationship is confirmed by regression analysis. Regressing the binary AI adoption indicator on the firm-level GenAI exposure measure, controlling for firm size, age, sector and year fixed effects, yields a positive and statistically significant coefficient, indicating that higher occupational exposure to generative AI is a strong predictor of AI adoption.

⁴⁵ A further limitation comes from the fact that market concentration measures themselves rely on several data sources, for which joint availability is required.

⁴⁶ More precisely, the first sample is based on all patent of invention applications of kind A (national), while PCT filings are evaluated using standardised filing requirements that make these applications more uniform across countries.

⁴⁷ Calligaris et al. (2024^[57]) innovate with respect to the existing literature by using more narrowly defined industries than previous cross-country studies, going from the 2-digit to mostly 3-digit level. In addition, it innovates by accounting for the international dimension of competition when defining markets, relying on a taxonomy of industries that defines whether each industry competes mainly at the domestic, European, or global level. The countries included in the sample are Belgium, Denmark, Finland, France, Germany, Greece, Hungary, Italy, Norway, Poland, Portugal, Slovenia, Spain, Slovenia, Sweden, and the United Kingdom for the domestic and European buckets, while in the global one also Japan, Korea, and the United States are included. For industries classified as competing at the European or global level, there is only one market (and, thus, one observation) per year, whereas for industries competing at the domestic level, concentration is measured for each country and year separately. The period covered is 2000-19.

⁴⁸ The analysis is constrained by a small sample size. The number of observations (industry-geographical market-year) having both data on market concentration and on AI inventions is 714. Moreover, to make the comparison meaningful, only observations with at least 4 AI-related patents were included in the analysis. This choice restricts the sample to 170 observations and is made to exclude observations for which concentration of patents is mechanically equal to one.

⁴⁹ The geographical region is identified by countries for domestic industries, while it takes value European or global for the tradable industries according to the taxonomy by Calligaris et al. (2024^[66]).

⁵⁰ The analysis is restricted on firms having both patents (included in PATSTAT) and enough financials information to compute their markups and it is based on a sample covering about 800 000 firms over 21 countries (Austria, Belgium, Czech Republic, Denmark, Estonia, Finland, France, Germany, Hungary, Ireland, Italy, Latvia, Luxembourg, the Netherlands, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, and the United Kingdom). It is worth noting that, given data limitations regarding firm financials, three countries – Korea, Japan, and the United States – where AI-related patenting activity is particularly relevant are excluded from the sample.

⁵¹ Broad ICT sectors include Manufacturing of computer, electronic and optical products (26), Publishing, audiovisual and broadcasting activities (58 to 60), Telecommunications (61), IT and other information services and (62 to 63), with numbers in parentheses identifying 2-digit industries in Nace Rev.2 classification.

⁵² The picture is suggestive and should be considered with some caution, as the low number of firms related to AI-related patent filings makes the lines somehow erratic, although the results are confirmed by regression analysis.

⁵³ While the literature on markup estimation highlights that markup levels are biased when recovered from financial data, (De Ridder, Grassi and Morzenti, 2026^[83]) show that the bias is common to all firms included in the production function estimation, usually estimated at industry level. Therefore, the bias does not affect studies looking at markup's dispersion across firms or trend over time. In the specific setting of this paper, the inclusion of industry fixed effects should attenuate this concern in the analysis.

⁵⁴ As discussed in footnote 53, column (1), while capturing a purely descriptive relationship, might report a biased measure of markups. The inclusion of industry fixed effects, specifically in our preferred specification (column (4)), should alleviate the bias as suggested by (De Ridder, Grassi and Morzenti, 2026^[83]).

⁵⁵ In the AI-markups relationship, causality might run in the other direction: AI inventions might be sustained by firms having higher markups before the innovation takes place.

⁵⁶ Further examining the cross-country heterogeneity in funding flowing to AI-related start-ups, Figure A B.5. Total VC funding to GenAI start-ups by region suggests that – based on the data available – a considerable share of this funding appears to be directed towards US-based start-ups.

⁵⁷ Since the variable is included as log, it is considered as $\log(1 + \text{total patents at the time of first VS round})$ to include all non-patenting firms – for which otherwise the log would not be defined.

⁵⁸ Examples of deals outside the traditional venture capital stages are “Debt Financing”, “Grant”, “Post-IPO Debt”, “Private Placement”, and other non-equity or hybrid instruments.

⁵⁹ The acquisition premium for Gen AI start-ups increases by 1.6 p.p the baseline likelihood of start-up acquisition across AI, which is 7.58%. This means that Gen AI have 21% higher likelihood of being acquired: $1.6 \text{ p.p} / 7.58\% \approx 21.1\%$.

⁶⁰ The other main exit route for high-growth start-ups is an IPO, which typically allows firms to scale while remaining independent competitors. To assess whether AI-related start-ups are generally more likely to experience *any* successful exit, a parallel linear probability model is estimated with an IPO indicator as the dependent variable. The results show that both non-GenAI and GenAI start-ups are *less likely* to exit via IPO relative to non-AI start-ups. This indicates that AI-related start-ups are not systematically more likely to experience positive exit outcomes overall; rather, their higher exit likelihood is specific to acquisitions. This distinction is important from a competition perspective, as acquisitions – unlike IPOs – may limit the emergence of independent competitors.

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Annex A. Technical appendix

Further details on markups

Defining markups from firms' cost minimisation

Following De Loecker and Warzynski (2012^[81]), which build on Hall (1988^[82]), the expression for markups is derived from the first order condition of the cost minimisation problem faced by firms with respect to an input of production free of adjustment costs (i.e., firms choose this input statically period by period, without intertemporal considerations) and for which firms are price takers.

Assume each firm f at time t follows a production function that converts inputs of labour L_{ft} , materials M_{ft} , and capital K_{ft} into output Y_{ft} , with a firm-specific productivity given by Ω_{ft} . Materials is the variable input purchased at price P_{ft}^M . Capital K_{ft} is a stock variable that firms choose dynamically (i.e., they do not choose it by period but before the present period). The production function can be written in its general form as:

$$Y_{ft} = Y(M_{ft}, L_{ft}, K_{ft}, \Omega_{ft})$$

Equation 7

Optimising firms choose their variable input, materials and labour, to minimise costs subject to achieving a given level of output. The first order condition with respect to the variable inputs, for instance, materials, emerging from the minimisation cost problem, leads to the condition:

$$P_{ft}^M = \lambda_{ft} \frac{\partial Y_{ft}(\cdot)}{\partial M_{ft}},$$

Equation 8

Where λ_{ft} is the Lagrange multiplier, measuring marginal costs (it is indeed is the marginal value of the objective function as we relax the output constraint). By rearranging λ_{ft} and multiplying both sides of the above expression by the firms' price P_{ft} , the right-hand side can be rewritten in terms of the output elasticity of materials multiplied by its inverse revenue share. In addition, markup μ_{ft} is given by price over marginal cost, $\mu_{ft} = \frac{P_{ft}}{\lambda_{ft}}$. Thus, the previous expression can be written as:

$$\mu_{ft} = \left(\frac{\partial Y_{ft}(\cdot)}{\partial M_{ft}} \frac{M_{ft}}{Y_{ft}} \right) \frac{P_{ft} Y_{ft}}{P_{ft}^M M_{ft}} = \frac{\alpha_{ft}^M}{S_{ft}}.$$

Equation 9

The markup is equivalent to the elasticity α^M of the flexible input M over the share of that flexible input's cost on revenues, S_{ft} . Stated differently, markups are given by the ratio between the output elasticity of the variable input chosen and its revenue share. Note that the output elasticity will depend on the specific production function adopted.

Estimation of markups

Markups can be estimated using balance sheet data on revenues and input spending. The methodology relies on estimating in each 3-digit industry the output elasticity of the variable input α_M . The advantage of

the method is that it does not require assumptions on the demand side or on how firms compete. Instead, it requires two key assumptions to hold. First, at least one input of production can be adjusted without frictions. Second, firms are price takers for that input and produce by minimising their costs.

Building on standard assumptions in the literature (De Ridder, Grassi and Morzenti (2026_[83])), the following Cobb-Douglas production function is assumed in each 3-digit industry:¹

$$Y_{ft} = \Omega_{ft} M_{ft}^{\alpha_M} L_{ft}^{\alpha_L} K_{ft}^{\alpha_K}$$

Equation 10

Taking logs of both sides (and using lowercase variables to represent the log of the original variable), the previous relationship can be rewritten as follows:

$$y_{ft} = \alpha_M m_{ft} + \alpha_L l_{ft} + \alpha_K k_{ft} + \omega_{ft}.$$

Equation 11

Following De Loecker and Warzynski (2012_[81]), a control function approach –and, in particular, the Akerberg et al. (2015_[84]) methodology– is used to estimate econometrically the parameters of the production function. Following the literature, materials are assumed to be the flexible input, given that labour is likely to face higher frictions like hiring and firing costs and capital is usually set dynamically and lasts for multiple periods.²

This procedure therefore provides an estimate of the elasticity of all inputs, including elasticity of materials, $\hat{\alpha}^M$, in each industry. Since cost of materials as a share of the firm's revenue (S_{ft}^m) is directly available in the data, the estimated markups are given by:

$$\hat{\mu}_{ft}^M = \frac{\hat{\alpha}^M}{S_{ft}^m}.$$

Equation 12

In the baseline specification, the industry-specific production functions are estimated separately for each 3-digit industry but are constant over time. The specification with constant elasticities over time is the preferred one for various reasons. First, from a theoretical point of view, the equilibrium does not vary with time but is static. Note that, despite the 10-year period analysed, longer periods have been used in other papers in the literature. More importantly, the time span covers the great financial crisis (GFC), and the elasticities computed using the years of the GFC would not only capture technological changes in the production function but would also be affected by the economic cycle. While the varying material inputs share in the denominator of the markup calculation accounts for seasonal variations in markups, output elasticity in the numerator should purely represent the technology of the industry. From a more practical point of view, the Orbis dataset has relevant differences in the sample composition over the years, with the sample covered by the database increasing significantly in more recent years. Finally, and importantly, to estimate the production function, a minimum number of observations in each industry is needed. Thus, a trade-off between going very detailed in terms of industries and splitting the sample in multiple periods arises. For the purposes of this exercise and for comparability with concentration measures, the highest granularity possible in terms of industries has been preferred. However, pooling all years within an industry implies that technology is assumed to be constant over time in each 3-digit industry. As the sample covers two decades, it is possible that technology instead varied. Therefore, in a robustness check, time varying production functions are estimated, allowing for output elasticities to change over time. To overcome the lack of sufficient observations, recent work has implemented the strategy of considering rolling windows between 3 and 10 years (see Traina (2018_[85]) and Baqaee and Fahri (2020_[86])). Thus, in a robustness exercise at the end of this Annex, the following alternative production function estimations have been carried out: i) estimations for each 5-year rolling window at the NACE 2-digit level (no year fixed effect) (cf.

Baqae and Farhi (2020^[86]) and De Loecker et al. (2020^[45]); ii) pooled estimations at the NACE 2-digit level (no year fixed effect); iii) 6-year rolling windows at the NACE 3-digit level (no year fixed effects). The different combinations of time window and industry level of disaggregation (6 years at 3-digit level versus 5 years at 2-digit level) have been adopted to meet the criteria on the minimum coverage in the sample to implement the estimation procedure. Note that the elasticities with rolling windows are computed in each year on a different sample of firms and are affected by the sample composition besides technological change. Finally, pooled estimation at the NACE 2-digit level has been reported since it has been used as a methodology by several references in the literature (De Loecker, Eeckhout and Unger (2020^[45])).

The methodology used in this study has been criticized for not being able to purge the measurement error resulting from the use of revenue instead of output quantities in the first stage of the production function estimation. As discussed in Klette (1996^[87]), De Loecker and Warzynski (2012^[81]) and Bond et al. (2021^[88]), the estimation of markup levels according to this methodology is biased downwards when deflated revenues are used, as in this case, instead of output quantities (not available in the data). However, De Ridder et al. (2026^[83]) point out that the dispersion and trends over time of markups are unbiased, and their correlations with other variables are still informative. They benchmark markups estimated with revenue data against markups estimated with quantity data. They argue that although the estimated markups under this methodology may be not accurately informative about the level of the true markups because of the biases in the elasticities, they can still provide informative results about their dispersion and time series trends. Therefore, the focus in this work is on time series trends of markups and their correlations with other economic information rather than on the levels themselves.

¹ De Ridder et al. (2026^[83]) show in a simulation exercise that a Cobb-Douglas production function matches the markup dispersion closer than a translog production function when revenue data is used.

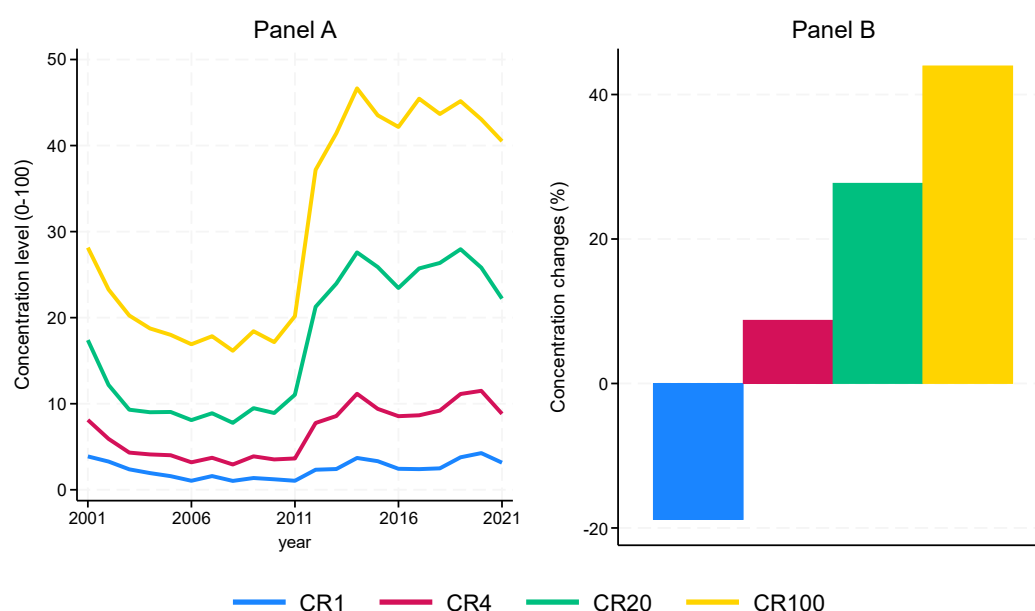
² Outliers in raw materials are removed by trimming the largest 1% of observations in each year of the sample. Once markups are estimated, outliers are removed by symmetrically trimming 3% of observations, as standard in the literature.

Annex B. Further graphs and tables

AI development and competition

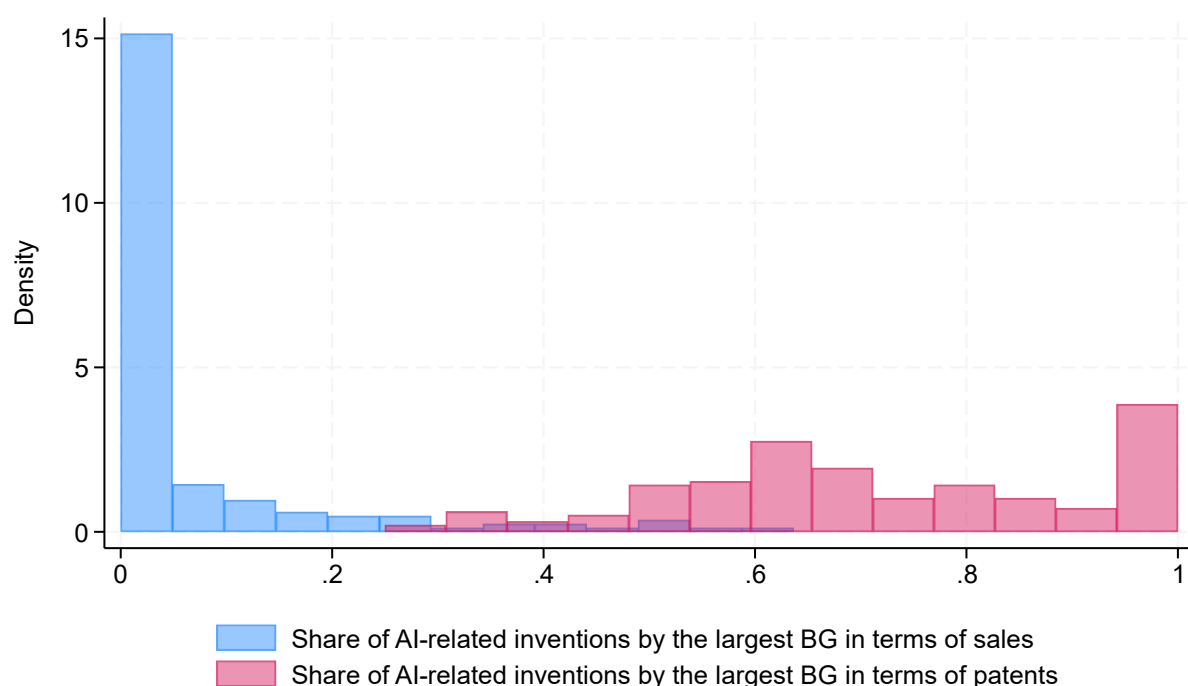
Concentration

Figure A B.1. Country share

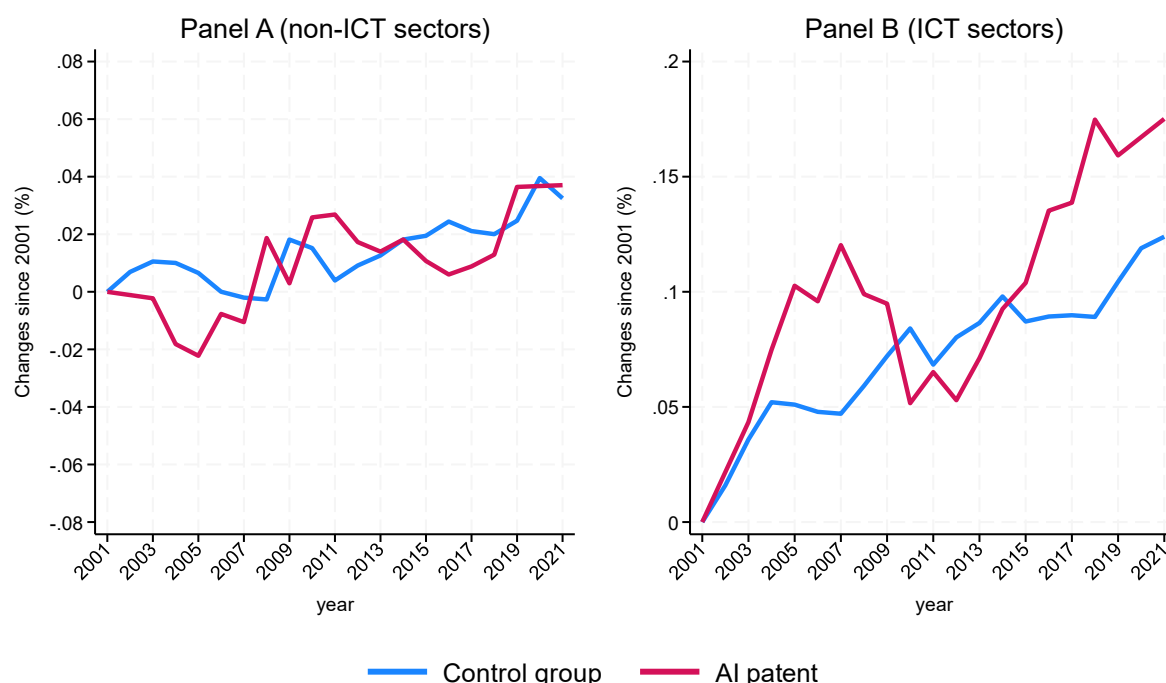


Note: In Panel A, the lines represent the AI-related patent concentration ratio, calculated as the share of patents associated respectively, the largest single one (CR1, blue), the top 4 (CR4, red), the top 20 (CR20, green), and the top 100 (CR100, yellow). In Panel B, the bars represent the growth rate, between 2001 and 2021, of each indicator of Panel A. The graphs are based on Patents of Invention applications filed under the Patent Cooperation Treaty, contained in PATSTAT, and associated with the AI tag developed by OECD (2025^[60]).

Source: OECD calculations based on PATSTAT Global and Moody's Orbis.

Figure A B.2. Distributions of patenting concentration

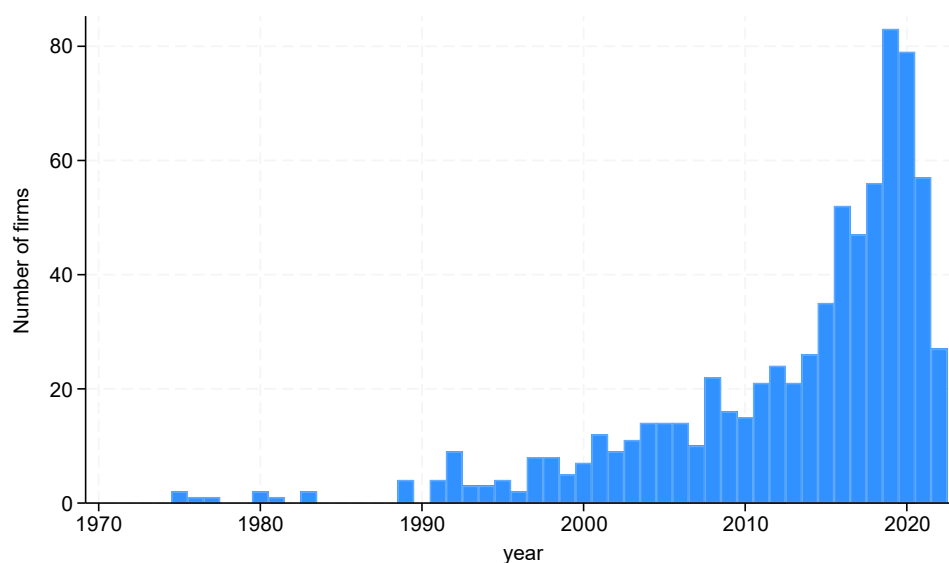
Note: The two histograms represent the distribution, across markets (industry-geographical region pairs) of two variables capturing concentration in patenting activity. The red histogram represents the share of AI-related inventions associated to the largest business groups in terms of patents (patent leaders in short). The blue one represents the share of AI-related inventions owned by the largest business groups in terms of sales (patents to market leader in short). This sample includes 15 industries (mostly from Manufacture of basic pharmaceutical products; Manufacture of computer, electronic and optical products; Professional, scientific, and technical activities) over the years 2000-2019. Each industry is classified either as domestic, European, or global, according to the taxonomy by Calligaris et al., (2024^[66]). The countries included in the sample are Denmark, Finland, France, Germany, Greece, Hungary, Italy, Norway, Poland, Portugal, Slovenia, Spain, Sweden, and the United Kingdom for the domestic and European buckets, while in the global one also Japan, Korea, and the United States are included. Source: OECD calculations based on PATSTAT Global and Moody's Orbis.

Markups**Figure A B.3. Markups of AI patentees across AI intensity taxonomy**

Note: The lines in the figure represent the growth rate of average markups across two groups of firms. The blue line is based on a sample of patenting firms (non AI-related), while the red one is based on firms associated with at least one AI-related invention, and only for the years after the first AI-related invention. The sample is further split into sectors considered as broadly related with ICT (Manufacture of computer, electronic and optical products (26), Publishing, audiovisual and broadcasting activities (58 to 60), Telecommunications (61), IT and other information services and (62 to 63), Panel B) and other sectors. Countries included in the sample are Austria, Belgium, Czech Republic, Denmark, Estonia, Finland, France, Germany, Hungary, Ireland, Italy, Latvia, Luxembourg, the Netherlands, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, and the United Kingdom.

Source: OECD calculations based on PATSTAT Global and Moody's Orbis.

Figure A B.4. AI switchers

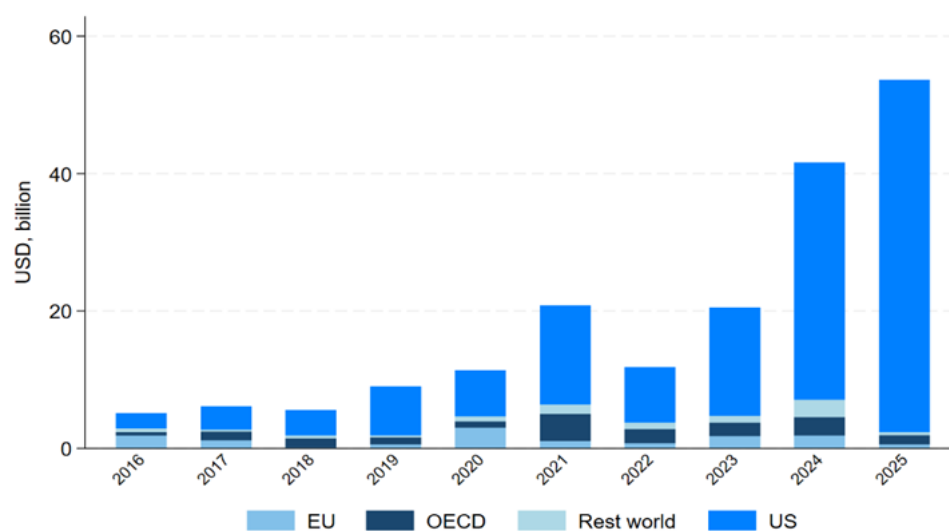


Note: The figure represents the number of firms in the sample (having data for both markups and inventions) that each year are associated for the first time with an AI-related invention. Countries included in the sample are Austria, Belgium, Czech Republic, Denmark, Estonia, Finland, France, Germany, Hungary, Ireland, Italy, Latvia, Luxembourg, the Netherlands, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, and the United Kingdom.

Source: OECD calculations based on PATSTAT Global and Moody's Orbis.

AI start-up ecosystem – further results

Figure A B.5. Total VC funding to GenAI start-ups by region



Note: The figure reports the amount of VC equity financing expressed in USD billions channelled to GenAI start-ups in the OECD start-up database. EU includes the 27 member countries of the European Union, OECD includes other OECD member countries excluding EU countries and the US, reported on their own. Rest world includes countries not included in the categories mentioned above.

Source: OECD calculations based on OECD Start-ups Database.

Competition in the age of AI

Initial evidence from microdata

No. 62

Artificial intelligence (AI) can reshape markets, yet its implications for competition remain underexplored. This paper develops a conceptual framework distinguishing AI users from developers, and generative (GenAI) from non-generative AI, showing how competitive mechanisms differ across these dimensions. Using multiple microdata sources, the analysis documents several findings. Adoption of non-GenAI is not associated with significant increases in market power. Descriptive evidence on firms' exposure to GenAI suggests opportunities for smaller firms alongside advantages for firms with stronger existing capabilities. Concentration in AI innovation is correlated with higher sales concentration. AI-related patenting is associated with faster markup growth, particularly in the ICT sector, where AI is an output. The AI start-up ecosystem is dynamic and attracts substantial venture capital, but start-ups are frequently acquired by large incumbents. Overall, the evidence points to a dynamic yet uneven landscape, underscoring the need for continued monitoring as AI diffusion progresses.